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NEUROHORMONAL REGULATION OF
DROSOPHILA PERICARDIAL NEPHROCYTES

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MAGISTRALE



Ai miei genitori, pilastri della mia esistenza.

A Giuseppe, l'amore della mia vita.

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SUMMARY

Chronic kidney disease (CKD) has emerged as a significant contributor to illness and mortality in the 21st century. CKD, as defined by the guidelines set forth by Improving Global Outcomes, represents a persistent abnormality in kidney structure or function that endures for a period exceeding three months. Due to the intricate interplay between the kidneys and other bodily organs, CKD exerts a substantial impact on multiple systems. Given the intricate and multifaceted nature of the human renal system, both in terms of *in vivo* and *in vitro* studies, researchers have found it imperative to explore alternative avenues. In this pursuit, *Drosophila melanogaster*, commonly known as the fruit fly, has risen to prominence as a remarkable model organism for investigating renal diseases. This choice is rooted in the striking genetic similarities and shared characteristics between *Drosophila* and humans, making it an invaluable tool in unraveling the complexities of CKD. The excretory system of *Drosophila* includes Malpighian tubules (MTs, analogous to the renal tubular system) and nephrocytes (analogous to podocytes and proximal tubules). While the regulation of MTs' function in *Drosophila* is known to involve neurohormones such as Capa, Diuretic hormone (DH) 31, DH44, Leucokinin (Lk), and tyramine, whose receptors are expressed in the epithelial cells of MTs, the molecular mechanisms governing ion and water transport in *Drosophila* nephrocytes remain poorly understood. Pericardial nephrocytes, due to their unique location, are responsive to almost all systemic physiological changes. Single-cell transcriptomic analysis of nephrocytes has suggested the possibility that these cells could also be subject to neurohormonal control, given the presence of transcripts for neuropeptides and biogenic amines receptors. Based on this idea, this thesis addresses these knowledge gaps, by characterizing different neurohormone receptors in the pericardial nephrocytes, encompassing: 1) an anatomical screening of neurohormone receptors using the Gal4-UAS system; 2) calcium imaging with the CaLexA system to gain a deeper understanding of neurohormonal regulation; and 3) knocking down receptors using RNAi to assess their role in the excretion and osmotic homeostasis.

In *Drosophila*, there exist hormone-producing cells within both the brain and the ventral nerve cord (VNC). Recently, all the neurohormone-producing cells within the *Drosophila* brain have been identified. This thesis extends this investigation to the VNC. Specifically, we directed our attention towards neuroendocrine cells responsible for producing hormones Lk and DH44, known to exert influence on MTs. By leveraging the VNC connectome, we were able to identify putative neurohormone-producing cells that exhibited characteristics consistent with those expected in neurohormone-producing cells. Notably, their morphology closely resembled what had been observed through the lens of confocal microscopy, further enriching our understanding of this intricate system. In summary, this thesis provides a comprehensive anatomical and functional characterization of neurohormonal pathways regulating pericardial nephrocytes and also delved

into the intriguing question of how these cells function when these receptors are absent, shedding light on their role in maintaining bodily fluids. Moreover, this exploration extended to uncover the potential neurons within the VNC, which may have a crucial role in producing neurohormones. This facet of our research not only introduces new and promising avenues for understanding the regulation of the excretory system in *Drosophila* but also holds the potential to provide novel insights into this important osmoregulatory system.

INTRODUCTION

Osmotic homeostasis is a crucial physiological process that helps organisms maintain a stable internal environment by regulating the balance of water and solutes, which is critical for the normal functioning of cells, tissues, and organs. This process ensures that cells have the proper environment to carry out their functions and helps regulate bodily processes, including fluid balance, blood pressure, and waste elimination, ultimately contributing to overall physiological health and well-being (Danziger and Zeidel, 2015).

The kidneys are the primary organs with an important role in maintaining osmotic and ionic homeostasis. Nephrons are functional units of the kidneys that are differentiated into a filtration unit known as the renal corpuscle or glomerulus and a segmented tubular resorption compartment. The glomerulus is composed of 4 types of cells: glomerular endothelial cells, mesangial cells, parietal epithelial cells and podocytes, so called because of their elaborate foot-like projections. Podocytes form an intricate interdigitating pattern around the glomerular capillaries and neighboring podocytes are physically connected to their foot processes via slit diaphragm (SD), constituted by intercellular connections. SD serves as the principal urine filtrate exit point and is considered to be crucial in the selective retention of high-molecular-weight plasma elements. Primary urine filtrate collects within the Bowman's capsule and drains through a network of epithelial tubules that includes the proximal tubules, the loop of Henle, the distal tubules, and a final collecting duct. The nephrons' renal tubules and collecting ducts express different ion and water channels and transporters, which concentrate and change the composition of the urine filtrate through resorption and secretion (Scott and Quaggin, 2015). The kidney's primary function is to control the body's fluid, electrolyte, and acid-base balances to provide a stable environment for tissue and cell metabolism. Balancing solute and water transport, excreting metabolic waste products, preserving nutrients, and controlling acid-base balance keeps the organism alive (Wallace, 1998). When this control is lost, there is onset of diseases that can lead even to death, if not treated in a timely and effective manner.

Chronic kidney disease (CKD) has emerged as one of the main causes of sickness and death in the 21st century. CKD is defined by the Improving Global Outcomes (KDIGO) guidelines as the presence of an abnormality in kidney structure or function that continues for more than 3 months. This can include impairment in glomerular filtration rate; albuminuria; urinary sediment, histology or imaging abnormalities suggestive of kidney damage; renal tubular disorders; or previous kidney transplantation (KDIGO, 2013). Due to the crosstalk between the kidney and other organs, CKD has significant impact on multiple systems, including energy metabolism, inflammation, neuroendocrine signaling, bone homeostasis, and the gastrointestinal,

pulmonary, and nervous systems (Zoccali *et al.*, 2017). Based on the latest Italian census data, the prevalence of CKD at all stages was 6.3% in 2010 (age- and gender-standardized number). Early stages (I and II), which are characterized by abnormal albuminuria, are more common than stages III-IV (3.8 % vs. 2.5 % for both males and females). Nearly 8,000 people were studied and according to these numbers, Italy has 2.2 million estimated CKD patients, with 1.3 million in the precocial stages (I and II) and 860 thousand in stages III-IV (De Nicola *et al.*, 2011).

Given the intricate and challenging nature of studying the mammalian renal system both *in vivo*, owing to its complex anatomical structure, and *in vitro*, due to the intricate physiology of its components that necessitate coordinated functionality, it becomes essential for research efforts to shift towards simpler model organisms, such as *Drosophila melanogaster*.

***Drosophila melanogaster* as model for renal disease**

One of the most studied model organisms in biology, particularly in genetics, is *Drosophila melanogaster*. This is due to its traits, which include its small size (it can maintain huge number of mutants in small lab space) and short life cycle (it can be raised in a lab in about 2 weeks), high productivity (600 embryos produced per female fly), polytene chromosomes in the salivary glands of mature larvae, only having 4 pairs of chromosomes - 3 autosomes and 1 sexual - and no genetic recombination in males. Its genome has been sequenced, and the species experiences frequent genetic mutations. In addition, the powerful Gal4/UAS expression system was standardized for using in *Drosophila* ([Brand and Perrimon, 1993](#)), allowing the temporal and tissue-specific expression of any chosen gene or construct. Genetically, humans and *Drosophila* are highly similar; the gene pool of the fly contains around 65% of putative orthologs of all known hereditary disorders, and roughly 70% of *Drosophila* proteins have a mammalian homolog. This enables the use of this remarkable model organism to investigate the potential genetic basis of human diseases, including renal diseases ([Dow *et al.*, 2010](#); [Helmstädter and Simons, 2017](#), [Koehler and Huber, 2023](#)). *Drosophila* is a useful model to discover the genes responsible for renal disease ([Zhang *et al.*, 2013](#)); for researching and understanding monogenetic disorders ([Hermle *et al.*, 2017](#)), which are caused by single gene mutations, for example the *Drosophila*'s ortholog gene *cindr* made possible to study the pathogenic mechanisms underlying of *CD2AP* gene mutation in podocytes, monogenic risk factor for nephrotic syndrome in humans ([Fu *et al.*, 2017](#)); it is an excellent model system to carry out large scale *in vivo* drugs screenings ([Wang *et al.*, 2022](#)); a promising tool for the study of transporters ([Wang *et al.*, 2018](#)); a diet-induced diabetic model for studying the connection between high dietary sucrose and diabetic nephropathy ([Na *et al.*, 2015](#)).

Excretory system in *Drosophila*

Insects have analogs which perform same functions of glomerular kidney, since human tubule and glomerular podocytes have recognizable analogs in the insect's body cavity. The excretory system of *Drosophila* consists of MTs (renal tubular system's analog) and nephrocytes (analogs of podocytes and proximal tubule), which are anatomically separated from one other (Figure 1). However, both components are necessary for the *Drosophila*'s excretory system.

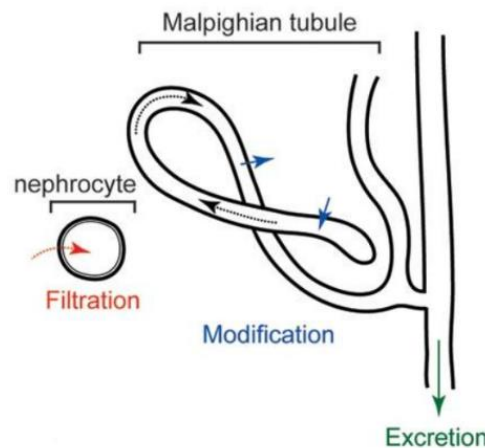


Figure 1: Schematic representation of *Drosophila* excretory system. [Reference image: Weavers H, Prieto-Sánchez S, Grawe F, et al. The insect nephrocyte is a podocyte-like cell with a filtration slit diaphragm. *Nature*. 2009;457(7227):322-326. doi:10.1038/nature07526 (modified)]

MTs' primary function is to detoxify hemolymph by eliminating waste metabolites and toxins and maintaining an appropriate balance of organic solutes and water. MTs are extremely permeable to low-molecular-weight molecules, providing an important function in actively uptaking and filtering catabolites, solutes (e.g., amino acids, sugars, ions), and water from hemolymph, producing primary urine. The soluble chemicals and water essential for the insect are finally reabsorbed in the proximal segment of the MTs, the rectum, or the ileum, keeping the internal osmotic pressure constant. MTs have the capacity to transform potentially harmful nitrogenous waste solutes generated through protein and purine metabolism into various end products. These end products may include uric acid, allantoin, allantoic acid, urea, or ammonia, with the specific conversion determined by the enzymes and biochemical pathways in operation. MTs can even actively excrete toxic substances of natural and chemical origin. In *Drosophila*, MTs additionally have a protective role, because they have the capability to detect the bacterial infections and produce various anti-microbial peptides, providing a cell-autonomous immune system (Farina *et al.*, 2022). The adult *Drosophila* possess four tubules (a pair on each side), which are organized

precisely and asymmetrically. A single MT has an inner luminal diameter of 17 μm , a length of around 2 mm, and can transport their own cell volume of fluid every 10 sec. The MTs can be divided into 3 genetically and physiologically distinct domains: initial, transitional, and main segments (Figure2).

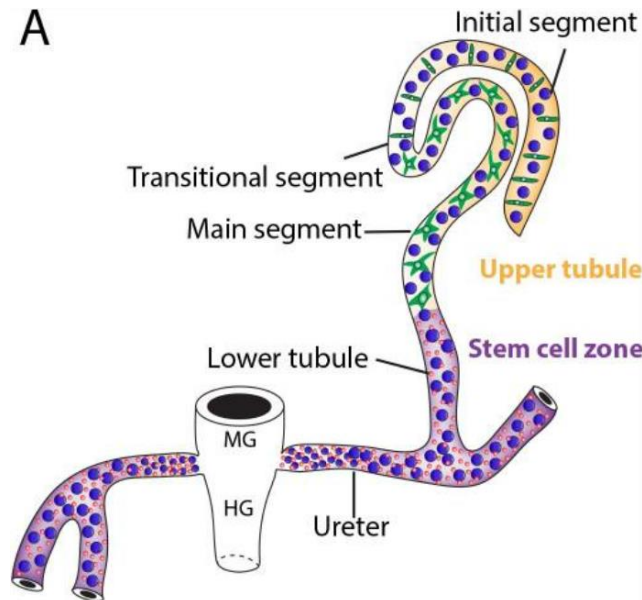
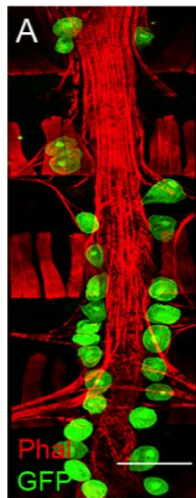


Figure 2: Illustration depicting the adult MT of *Drosophila*.

(Reference image: Wang C, Spradling AC. An abundant quiescent stem cell population in *Drosophila* Malpighian tubules protects principal cells from kidney stones. *Elife*. 2020;9:e54096. Published 2020 Mar 16. doi:10.7554/eLife.54096)

The main segment contains approximately 75 of the 100-150 cells that comprise each tubule. This region, which is made up of principal and stellate cells, is primarily responsible for the production of “primary urine” in *Drosophila* (Jung *et al.*, 2005). The principal cells are found in the main segment and possess basolateral Na^+ , K^+ , and Cl^- ion cotransporters, a Na^+ -dependent solute transporter, and a Na^+ -dependent Cl/HCO_3 exchanger. The Vacuolar-type H^+ -ATPase is responsible for ion transport at the apical cell membrane of the principal cell, where it pumps protons from the cell into the MT lumen and establishes the gradient required for secondary movement of Na^+ and K^+ from the hemocoel into the lumen, by the Na^+/H^+ and K^+/H^+ exchangers (Miller *et al.*, 2013). Stellate cells are largely important for chloride transport and aquaporin-mediated water movement, and they are found throughout the initial, transitional, and main segments of the posterior MTs (Dow *et al.*, 2010).

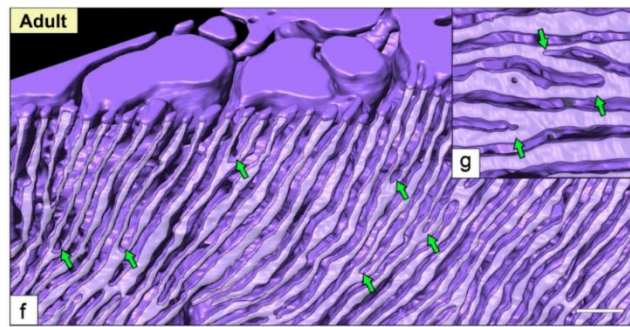
Unlike MTs, the molecular mechanisms behind ion and water transport in nephrocytes are not well known. Adult *Drosophila* nephrocytes are classified, depending on where they are located, as garland cells (binucleate) and pericardial cells (mononucleate). Garland cells can be seen between the salivary glands or close to the esophagus. Pericardial nephrocytes are situated on either side of the heart organized in two rows of 20-25 cells from the first through the sixth segments of the organ (Figure3) (Na *et al.*, 2013).



*Figure 3: Localization of pericardial nephrocytes along the heart. Adult heart tissue was visualized by staining with Phalloidin (in red). The expression of the dorothy (dot) reporter GFP was used to label the adult pericardial nephrocytes. Scale: 100 μ m. [Reference Image: Gera J, Budakoti P, Suhag M, Mandal L, Mandal S. Physiological ROS controls Upd3-dependent modeling of ECM to support cardiac function in *Drosophila*. *Sci Adv*. 2022;8(7):eabj4991. doi:10.1126/sciadv.abj4991 (modified)]*

The unique position of the pericardial nephrocytes makes them responsive to almost all systemic changes and influences in the fly body. Examples include changes in hemolymph composition (ions, nutrients, and pathogens), as well as changes in physiological parameters like pressure (Koehler and Huber, 2023).

Nephrocytes employ a combination of filtration, endocytosis, and processing mechanisms to control the composition of the hemolymph, allowing them to capture and metabolize particles. It is assumed that local pressure differences resulting from the movement of proximal peristaltic tissues (the heart and oesophagus) promote ultrafiltration and irrigation of nephrocytes, considering that *Drosophila* lacks a closed circulatory system (Denholm and Skaer, 2009). Within the nephrocytes, the endocytic cargo is either sorted for degradation by lysosomes or recycled back to the hemolymph. Unwanted cargo cannot be destroyed; however, it can be inactivated and permanently preserved intracellularly (Helmstädter *et al.*, 2017). These cells do not play a role in the production of urine. Nevertheless, the high endocytic activity can still be referred to as "excretory" activity. To maximize their effectiveness in this role, it is feasible to enhance their surface area by creating an intricate network of labyrinthine channels or lacunae. These structures result from plasma membrane (PM) infoldings and are enveloped by nephrocyte foot processes. Additionally, nephrocytes possess a highly specialized filtration junction known as the SD. The SD is comprised of either mono- or bi-filaments spanning narrow slit-like openings, measuring just 30 nm in width. Each nephrocyte is enveloped by a negatively charged basement membrane, contributing to its unique filtration properties (Figure 4) (Weavers *et al.*, 2009).



*Figure 4: Reconstructed surface of pericardial nephrocytes via FIB/SEM. Using their serial FIB/SEM images, the basal surface of pericardial nephrocytes was used to recreate the 3D structure of pericardial nephrocyte. Pictured in whitish purple is the foot processes' basal surface membrane, which adheres to the basement membrane. The foot processes are connected to one another in a parallel fashion by means of anastomoses at their ends (green arrows). Scale bar: 500 nm. [Reference Image: Kawasaki Y, Matsumoto A, Miyaki T, et al. Three-dimensional architecture of pericardial nephrocytes in *Drosophila melanogaster* revealed by FIB/SEM tomography. *Cell Tissue Res.* 2019;378(2):289-300. doi:10.1007/s00441-019-03037-3 (modified)].*

Filtrate gets endocytosed from the sidewalls of the lacunae via the SD and basal membrane, which act as a size- and charge-selective barrier because this barrier properly restricts out particles with a diameter greater than 10-12 nm and the basal membrane's intrinsic negative charge functions as a charge-selective filter to block the entry of molecules with a high negative charge (Denholm and Skaer, 2009).

Nephrocytes for glomerulopathies research

Nephrocytes and glomerular podocytes share strong morphological, molecular, and functional similarities, to the point that nephrocyte has been recognized as an invertebrate model that may be able to satisfy the prerequisites of an additional model for glomerular disease, representing a unique model system for having a functional SD in an extremely genetically tractable model organism. SD auto-cellular connections are generated by the orthologs of the mammalian transmembrane Ig-domain superfamily proteins nephrin (sticks and stones, sns) and NEPH1 [kin of irre, kirre; also known as dumbfounded, duf]. Both proteins have PM localization, similar to vertebrate podocyte. The barrier itself is established via interactions that occur between the extracellular domains of these proteins which can be homotypic and heterotypic. SD are placed to covering these invaginations, as being a roof for each lacuna ([Figure5A](#)); in tangential sections proteins pattern resembles fingerprints ([Figure5B](#)) ([Weavers et al., 2009](#)).

Vertebrate nephrin and NEPH1 establish a multi-protein complex with zonula occludens-1 (ZO-1), CD2AP, and podocin at the SD. Fly orthologues contribute to the SD, and *in situ* hybridization revealed that *pyd* (Polychaetoid, ZO-1 orthologs), *CG31012* (CD2AP), and *Mec2* (NPHS2/podocin) are expressed in nephrocytes. Utilizing yeast two-hybrid analysis of Sns and Duf intracellular domains, it emerged that Sns interacts with Mec-2 (podocin) while Duf interacts with Pyd (ZO-1). The intracellular domains' interaction of Duf and Pyd with these proteins was separately validated by co-immunoprecipitation ([Weavers et al., 2009](#)).

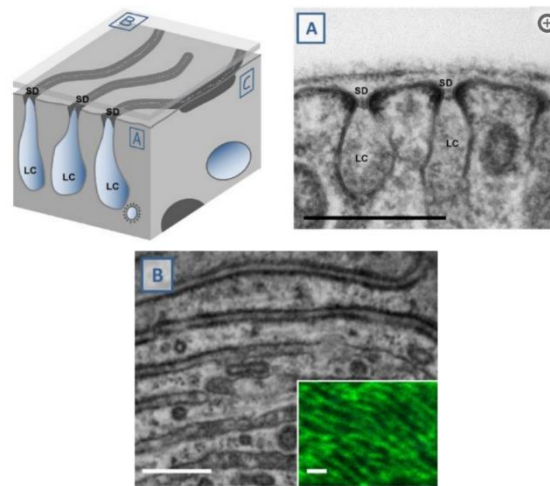


Figure 5: Nephrocyte surface observed via electron microscopy. A) Perpendicular section to the nephrocyte surface. The thin border that covers the entry point (near the electron density around the entry point) of the labyrinthine channels is the slit diaphragm. B) Surface's tangential section. The continuous lines are the slit diaphragm. Scale bar 500 nm. LC refers to labyrinthine channels, SD refers to slit diaphragm. [Reference image: Helmstädter M, Huber TB, Hermle T. Using the Drosophila Nephrocyte to Model Podocyte Function and Disease. Front Pediatr. 2017;5:262. Published 2017 Dec 7. doi:10.3389/fped.2017.00262 (modified)].

Development as well as preservation of the SD but also absorption of molecules into the labyrinthine channels for processing via endocytosis rely on Sns and Kirre. Using transgenic flies, RNA interference (RNAi) was conducted against Sns and Kirre. As demonstrated by transmission electron microscopy analysis, these RNAi significantly affected the number of SDs and, through the co-incubation of different size tracers it was also affect the capacity to distinguish the various sizes of molecules that penetrate cells. Thus, a smooth cell surface is produced by the lack of function of the SD proteins, which is comparable to the deletion of the podocyte foot process ([Zhuang et al., 2009](#)).

Endocytosis depends on a functional SD

Proper assembly of functional endocytosis in nephrocytes involves a diverse array of cellular components. Apical polarity regulators (Bazooka, aPK-C, PAR-6, and the Crb complex)

and the basolateral polarity module (Discs large, Scrib, Lethal giant larvae, and Par-1) play pivotal roles in SD formation, endocytosis, and hemolymph filtration (Heiden *et al.*, 2021; Mysh and Poulton, 2021). Disruption of basal proteins impacts the stability of *Drosophila* Crumbs protein, known to be present in dense SD structures, subcortical vesicles, and lacunae outlines. Notably, its intracellular domain interacts with Moesin, contributing to the endocytic process. Depletion of Crumbs results in reduced filtration rate, altered morphology, decreased SD density, and disturbances in endosomes/lysosomes, highlighting impaired vesicle trafficking (Hochapfel *et al.*, 2017). Crucial for vesicle trafficking and endocytosis are Rab GTPases in nephrocytes. Targeted silencing of Rab5, a marker for early endosomes, at different stages leads to the inward movement of fly nephrin. Acute silencing disrupts the endocytic clearance of ectopic nephrin, causing lateral dispersion of SD proteins within labyrinthine channels and ultimately architectural disintegration, emphasizing the significance of endocytosis in maintaining structural integrity (Lang *et al.*, 2022). Similarly, knockdown of Rabphilin, the *Drosophila* orthologue of RPH3A, prompts structural changes like the absence of labyrinthine channels, modified basement membrane, and notable effects on Rab3⁺; indicative of altered endocytic trafficking, molecular uptake, reduced endocytic vesicles and vacuoles, and downregulation of Cubilin expression (Selma-Soriano *et al.*, 2020). Cubilin and Amnioless are two proteins localized at the lacunae that work together as a receptor complex in nephrocytes. This complex is involved in removing pollutants from the hemolymph, as revealed by RNAi experiments of *CG32702* (Cubilin ortholog) and *CG11592* (Amnioless, AMN ortholog) genes. (Zhang *et al.*, 2013). Central to the maturation and sustenance of SD structure and nephrocyte function is the clathrin-mediated endocytosis (CME) pathway. Interference with CME pathway genes (e.g., Clathrin heavy and light chain, shibire, Adaptor Protein complex 2, Aux, Hsc70-4) through RNAi results in mislocalization of SD proteins, disruption of fingerprint-like patterns, and protein aggregates. *In vivo* protein uptake assays revealed compromised cell reabsorption capacity (Wang *et al.*, 2021). Inhibition of dynamin-mediated endocytosis through a temperature-sensitive mutant of *Drosophila* shibire impedes the lateral diffusion of ectopic nephrin, highlighting its significant influence on SD dynamics (Lang *et al.*, 2022).

The significance of cytoskeleton in regulating nephrocyte structure and function is evident in the suppression of four orthologs of human genes which are defective in monogenic SRNS (Steroid Resistant Nephrotic Syndrome), and involved in actin regulation (ACTN4, ARHGAP24, MYH9, ANLN). Knock down of these genes affects tracer uptake in nephrocytes (Hermle *et al.*, 2017). Particularly, suppression of Zip-KD (*Drosophila* ortholog of MYH9) leads to severe phenotypes, including significantly reduced fluorescent protein uptake and similar outcomes observed in silver nitrate (AgNO₃) sequestration assays by nephrocytes (Fu *et al.*, 2017).

Physiological importance of nephrocyte maintenance

Pericardial nephrocytes have distinct roles in addition to their primary function as a barrier. Positioned on both sides of the cardiac tube, they enclose it and facilitate non-cell autonomous processes that control the functioning of the *Drosophila* heart. Loss of pericardial nephrocytes in *Drosophila* can lead to cardiomyopathy that depends on the presence of Secreted Protein Acidic and Rich in Cysteine. However, this cardiomyopathy can be prevented through a nephrocyte-mediated clearance mechanism (Hartley *et al.*, 2016). Furthermore, nephrocytes influence cardiac physiology by regulating it through reactive oxygen species (ROS)-mediated paracrine signaling. Elevated ROS levels in pericardial nephrocytes, in a p38 dependent manner, modulate the function of neighboring cardiomyocytes. This regulation involves two septate junction proteins, Cora and Kune (Lim *et al.*, 2019). Furthermore, the ROS produced by pericardial nephrocytes serves as a metabolic signal necessary for supporting cardiac function by shaping the structure of the cardiac extracellular matrix (ECM) through the incorporation of optimal amount of Pericardin (Prc). This process is a result of communication between different organs: the physiological level of ROS in pericardial nephrocytes plays a crucial role in controlling the expression of *prc* in fat cells, which is essential for the development of the cardiac ECM in *Drosophila*. This is primarily accomplished by regulating the expression of the cytokine Unpaired 3 (Upd3) within the PCs. In turn, Upd3 released by the pericardial nephrocytes initiates JAK-STAT signaling in the fat cells, which is responsible for inducing *prc* expression (Gera *et al.*, 2022). In addition, nephrocytes are involved in preserving immune balance by preventing persistent activation of the Toll pathway. In terms of their mechanism, nephrocytes internalize Lys-type peptidoglycan from the body's circulation and direct it to lysosomes for breakdown, thereby averting local immune responses against the microbiota (Troha *et al.*, 2019). Therefore, pericardial nephrocytes can perform multiple functions essential to maintain organismal physiology and healthy lifespan.

Neurohormones and their signaling mechanisms

Hormones have a significant role in controlling these tissues, mainly because the central nervous system (CNS) does not directly innervate them. Neurohormones, comprising neuropeptides, peptide hormones, and biogenic amines, are produced by neurons and neuroendocrine cells situated in various locations, including the CNS, VNC, gut, and other peripheral tissues.

Neuropeptides are generated by activating certain genes that encode pre-pro-hormones or pre-pro-peptides, from which mature products can be released via enzymatic cleavage and further processed post-translationally.

Biogenic amines are commonly produced from specific amino acids such as tyrosine, tryptophan, histidine, and phenylalanine, which act as the precursors for their synthesis. These amines are generated within the body through enzymatic processes, notably decarboxylation. Following decarboxylation, the resulting amine can undergo additional enzymatic modifications to generate the various biogenic amines (Purves *et al.*, 2019).

Neuroendocrine neurons feature large cell bodies, where hormones synthesis takes place. The hormone-containing dense core vesicles are commonly transported to axon terminations or boutons, usually enlarged (Nässel and Zandawala, 2019).

Neurohormones signaling occurs on a slow timescale. These are frequently released at locations far from synapses or receptor sites. The expression of suitable receptors in target cells determines the precision of type of signaling and more diffuse communication. A significant number of neurohormone receptors are members of the GPCR (G-protein coupled receptor) family. This is a large and evolutionary conserved superfamily of integral cell membrane proteins that play a central role in signal transduction, specialized to receiving intercellular messenger chemicals (such as hormones, neurotransmitters, growth and developmental factors) as well as a variety sensory messages (such as light, odors, and gustative molecules) (Bockaert and Pin, 1999). GPCRs in *Drosophila* are divided into 4 families: rhodopsin-like, secretin-like, metabotropic glutamate-like, and atypical 7 transmembrane proteins. Commonly, the extracellular NH₂ terminus of GPCR is glycosylated, while the cytoplasmic COOH terminus is phosphorylated (Brody and Cravchik, 2000). 3 extracellular loops alternate with 3 intracellular loops to connect the “central core” of 7 hydrophobic α -helical domains that span the membrane and transduce signals through coupling to guanine nucleotide-binding regulatory proteins (G proteins). Via the catalysis of the GDP-GTP exchange on heterotrimeric G proteins (Ga-b γ), they regulate the synthesis of intracellular second messengers such as cAMP, inositol phosphates, diacylglycerol (DAG) and Ca²⁺ ions (Figure6) (Hauser *et al.*, 2006).

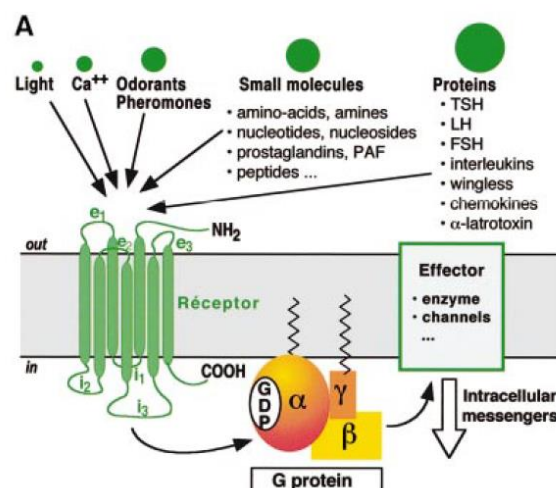


Figure 6: Schematic illustration of GRCP. GPCRs share a common core consisting of 7 transmembrane helices (TM-I to -VII) linked by 3 intracellular (i1, i2, i3) and 3 extracellular (e1, e2, e3) loops. The receptors can be triggered by a variety of messages and so that start the signal transduction. [Reference image: Bockaert J, Pin JP. Molecular tinkering of G protein-coupled receptors: an evolutionary success. EMBO J. 1999;18(7):1723-1729. doi:10.1093/emboj/18.7.1723]

Ca^{2+} is a ubiquitous intracellular signal that regulates local cellular response, and in multicellular animals also functions as a signal for coordinating systemic reactions at the whole-body level. The basal cytosolic Ca^{2+} levels, usually, are very low (10^{-7} mM) and are modulated by a powerful interplay of various Ca^{2+} transport systems located on the surface PM and intracellular compartment membranes, many of them operate as internal Ca^{2+} storage. The presence of a variety of Ca^{2+} binding proteins also influences free cytosolic Ca^{2+} concentration, a few integrate changes in cytosolic Ca^{2+} concentration with downstream effectors such as Ca^{2+} activated kinases, phosphatases, and transcription factors (Chorna and Hasan, 2012). Ca^{2+} elevation can govern a wide range of functions due to the Ca^{2+} signaling mechanism's adaptability in terms of speed, amplitude, and spatial-temporal patterning. The signaling cascade starts when a stimulus activates the Ca^{2+} signaling network, which produces a variety of Ca^{2+} -mobilizing signals. The latter activate the Ca^{2+} -feeding ON pathways in the cytoplasm. Then Ca^{2+} acts as a messenger to promote a variety of Ca^{2+} -sensitive activities. Finally, the OFF mechanisms, which are made up of pumps and exchangers, remove Ca^{2+} from the cytoplasm to return the cell to its resting state. Because many of the molecular components have many isoforms with something different properties, each cell type can take advantage of this diverse repertoire to build adaptable Ca^{2+} signaling networks (Berridge *et al.*, 2000). Synaptic transmission in the nervous system is the most advanced mechanism for Ca^{2+} -dependent systemic communication, in which neurotransmitters trigger Ca^{2+} channels to initiate electrical transmission, via direct passage of ions between the two nerve cells, or chemical transmission, through neurotransmitter release by synaptic vesicles (González-Aguilar, 2000).

Neurohormones influencing Malpighian tubules and excretion

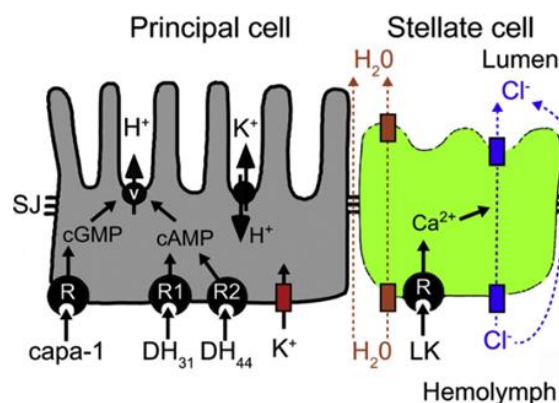
Neurohormonal signaling exerts significant control over the majority of *Drosophila* physiology and behavioral aspects.

Neuropeptides have roles in almost every part of daily life, for example one of the most studied is Pigment Dispersion Factor (PDF) peptide that controls clock function and sleep. Other aspects such as olfaction, taste, foraging, nutrition, aggression, mating, reproduction, learning are

regulated by these neuromodulators, as well as stress responses, fecundity, lifespan, development, growth, water balance and even metabolic function as demonstrated adipokinetic hormone (AKH) that regulates both carbohydrate and lipid mobilization. Currently, *Drosophila* has approximately 50 genes that encode neuropeptide precursors, and approximately same number of genes that encode peptide GPCRs (G-protein-coupled receptors) ([Nässel and Zandawala, 2019](#)).

On the other hand, dopamine, serotonin (5-HT), octopamine (OA), tyramine, and histamine are the most common monoamine neurotransmitters in *Drosophila*. In a complementary manner, these chemicals affect life processes like as learning and memory, circadian rhythms, aggression, mating behavior, thermoregulation, and a variety of other homeostatic mechanisms. In particular, 5-HT is involved in walking, olfactory learning and memory, and it is a sleep modulator. It also regulates the entrainment of circadian behavioral rhythms by influencing the molecular response to light, is involved in social motivation and sociability, and is associated with less aggressive behavior. Similarly, OA affects baseline locomotion, both appetitive and aversive learning, egg-laying, egg-retention and release. It even regulates starvation-induced hyperactivity, and sleep patterns (Rosikon *et al.*, 2023).

Several neurohormones have been shown to influence the activity of MTs that regulate water and ion homeostasis. In *Drosophila*, osmoregulation is affected by four neuropeptides such Capa, Lk, DH 31, DH44, and the biogenic amine tyramine; and it seems to work according to a two-cell model. Capa, DH31, DH44 have influence on principal cells since their receptors are expressed in these cells, while Lk regulates the activity of stellate cells because its receptor has been shown to localize only in stellate cells in MTs (*Figure7*). Tyramine, on the other hand, operates via a distinct mechanism because of its paracrine source, originating within principal cells and exerting its effects on related receptor in stellate cells. (Halberg *et al.*, 2015).



*Figure 7: A schematic illustration of the activities of diuretic hormones in adult *Drosophila*. Aquaporin channels are represented by orange rectangles, chloride channels by blue rectangles, and Kir potassium channels by red rectangles. SJ stands for septate junction, and V stands for V-type ATPase. [Reference image: Nässel DR, Zandawala M. Recent advances in neuropeptide signaling in *Drosophila*, from genes to physiology and behavior. Prog Neurobiol. 2019;179:101607. doi:10.1016/j.pneurobio.2019.02.003 (modified)]*

Over the years, researchers have conducted studies to investigate the mechanisms of action of these modulators.

The Capa peptides are members of the PRXamide peptide family, related to neuromedin U signaling in vertebrates (Terhzaz *et al.*, 2012). *Drosophila* capability (*capa*) gene encodes a pre-pro-hormone that is processed to produce Capa-1 and Capa-2, which are periviscerokinins, and Capa-3 (or pyrokinin 1), a member of the pyrokinin family. All Capa family members induce fluid secretion in *Drosophila* tubules, with Capa-1 being the most effective stimulator. These induce increase in intracellular Ca^{2+} concentration within the principal cells. Griess assay demonstrated the role of these peptides in increasing MTs nitrite production (Kean *et al.*, 2002). Fluid transport is stimulated by Capa by activating NOS (nitric oxide synthase) and downstream NO-cGMP signaling cascade components. dNOS is expressed in *Drosophila* MTs, and Capa stimulates this cascade through NOS activity in Capa-stimulated tubules (Davies *et al.*, 1997).

Patch-clamp analysis revealed that the tubule apical membrane has a tiny relative area that is strongly enriched in "maxi"-chloride channels, which coincides with stellate cell localization. Expression of a UAS-aequorin transgene in stellate cell for Ca^{2+} measurement, demonstrated that Lk raises Ca^{2+} quickly and specifically in stellate cells, in line with these cells' major role in regulating chloride conductance, resulting in a significant increase in fluid secretion rate (O'Donnell *et al.*, 1998). Fluid transport induced by Capa and Lk required a PLC-dependent signaling pathway, as proven by *norpA* (no receptor potential A) H52 and *norpAP24* mutants (which had lower PLC activity) that suppressed Capa- and Lk-induced fluid secretion. Because they work through rely on PLC, there is a rise in inositol trisphosphate (IP3) in tubules, both Capa and Lk act through via intracellular Ca^{2+} . Capa, which acts only on principal cells, increases IP3 levels in intact wild-type tubules. Cytosolic Ca^{2+} measurements in wild-type and *itpr* tubules (mutant for IP3 receptor) stimulated with Capa using targeted aequorin, revealing a biphasic rise of intracellular Ca^{2+} in wild-type but not in mutants, showing that IP3 receptor-mediated Ca^{2+} release from intracellular reserves contributes considerably to Ca^{2+} signaling and, as a result, diuresis in primary cells. Same observations were obtained for Lk-stimulated Ca^{2+} signaling and fluid transport in stellate cells as *Drosophila* S2 cells transfected with the Lk receptor have been

shown to have increased cytosolic intracellular Ca^{2+} and IP3 content after Lk stimulation. ([Pollock et al., 2003](#)).

In *Drosophila melanogaster*, the Dh44 gene encodes a Corticotropin-releasing factor-like 44-amino-acid peptide (DH44). This peptide is active on the MTs, stimulating fluid secretion at concentrations equal to or more than $10^{-7} \text{ mol l}^{-1}$, with cAMP responsible for this effect. Immunocytochemistry with an anti-cAMP antibody revealed that the levels were exclusively elevated in principal cells, with no influence on cGMP or Ca^{2+} levels. After activating the tubules with DH44, cAMP- and cGMP-directed phosphodiesterase was evaluated, and it became evident that stimulation increased cAMP (but not cGMP) phosphodiesterase levels. Thus, application of DH44 directly stimulates the enzyme that degrades its second messenger ([Cabrero et al., 2002](#)).

DH31 is a 31-residue amidated peptide in *Drosophila*, that is the homologue of vertebrate Calcitonin gene-related peptide. It showed to have diuretic activity on MTs, with an EC_{50} of 4.3 nmol l^{-1} . Stimulation with 50 nmol l^{-1} DH31 increased the lumen-positive transepithelial potential in the principal secretory segment of the tubule and resulted in a dose-dependent increase in cAMP production. Hyperpolarization was attributed to activation apical V-ATPase via cAMP in principal cells, the increase in transepithelial potential establishes a favorable electrochemical potential gradient for Cl^{-} to enter the lumen, determined its effect on the pH and K^{+} concentration of secreted urine ([Coast et al., 2001](#)).

Tyramine is a biogenic amine resulting from decarboxylation of the amino acid tyrosine. When tested on MTs through transepithelial potential variations, tyramine produced a detectable response at 1 nM and a maximal response at $\sim 100 \text{ nM}$. Tyramine-induced depolarizations were also associated with an increase in chloride conductance (near 0 mV), proved by presence tyramine caused a significant increase in transepithelial chloride conductance, due to higher chloride diffusion potential. Urine secretion rates in isolated tubules bathed in standard bathing medium were examined where there was added 100 mM yohimbine (a specific blocker of tyraminerbic signaling in the tubule), leading in a 60% rate reduction. Immunohistochemistry using a tyramine antibody revealed that principal cells were the primary location of tyramine synthesis ([Blumenthal, 2003](#)). *Drosophila* tyramine applied to tubules transgenic for aequorin::GFP fusion in stellate cells, for measurement of intracellular Ca^{2+} , it was observed a prominent, rapid Ca^{2+} rise. The tyramine response was concentration-dependent, with an EC_{50} of $1.77 \times 10^{-8} \text{ M}$. The concentration dependence of tyramine-induced intracellular Ca^{2+} increase was consistent with an effect on chloride shunt conductance. *Drosophila* tyramine signals through PLC and IP3 that elevate intracellular Ca^{2+} , and to trigger a massive and rapid increase in the chloride shunt conductance, as tested using mutants for PLC, norpA, and for *itpr* gene. ([Cabrero et al., 2013](#)).

In contrast to such detailed knowledge about MT regulation, studies examining the functions as well as the regulation of pericardial nephrocytes are largely lacking. However, a more profound understanding of these cells and their regulatory processes has emerged, attributed in part to the utilization of innovative sequencing techniques like single-cell transcriptomics. The gene expression that came from these analyses of nephrocytes prompted the idea that these cells could also be regulated in their function via neurohormonal control, due to the presence of several typical receptor transcriptomes that the corresponding receptors respond to neuropeptides or biogenic amines (*Figure8*, Li *et al.*, 2022).



Figure 8: Single cell-transcriptome analysis has identified the potential presence of neurohormone receptors in pericardiac nephrocytes (*Reference image*: Li H, Janssens J, De Waegeneer M, et al. *Fly Cell Atlas: A single-nucleus transcriptomic atlas of the adult fruit fly*. Science. 2022;375(6584):eabk2432. doi:10.1126/science.abk2432)

AIMS OF THE THESIS

In this thesis, we set out to achieve two broad aims:

- I. Characterization of neurohormone receptors in pericardial nephrocytes.
- II. Identifying putative neurons which produce neurohormone in *Drosophila* VNC.

Characterization of neurohormone receptors in pericardial nephrocytes

Three different types of analysis were conducted to accomplish the first aim.

Anatomical screen of neurohormone receptors: expression analysis was performed to identify potential neurohormone receptors expressed in pericardial nephrocytes. From the list of receptors came from single-cell transcriptome data, Gal4-UAS system was used to analyse the gene expression. This system enables the selective and temporal expression of specific genes or constructs.

Functional analysis as Calcium Imaging: A Calcium Imaging technique was employed to examine potential variation in intracellular Ca^{2+} levels when dissected nephrocytes were incubated with different ligands. The Ca^{2+} indicator CaLexA was used to analyze the response of nephrocytes to different neurohormones.

Knock down of positive receptors during the screening: RNAi was employed to knock down the receptors analyzed in the previous experiments. This utilized the well-established Gal4/UAS system (Dietzl *et al.*, 2007). Loss of function approach aimed clarify the effects of lack of receptor on nephrocyte function, to understand the role of these neurohormone receptors in maintaining fluid balance. These were assessed through water content measurements and defecation assay.

Identifying putative neurons which produce neurohormone in Drosophila VNC

Recently, neurohormone-producing cells have been identified in the *Drosophila* brain (Reinhard *et al.*, 2023, pre-print). As mentioned above, VNC also produces and releases neurohormones. Thanks to advancements in electron microscopy and deep learning, it is now possible to rebuild a connectome, a description of neurons and their connections, for large volumes of neural tissue.

Therefore, another goal was to identify the putative neurohormonal neurons in the female VNC using neuPrint+, an open access database and analysis ecosystem that organizes connectome data to facilitate biological discovery ([Plaza *et al.*, 2022](#)).

MATERIALS AND METHODS

Fly Lines, Husbandry and Crossings

All flies were maintained on standard cornmeal/molasses/yeast diet.

Oct-alpha-2R-T2A-Gal4 (#84610, Bloomington), 5HT7-T2A-Gal4 (#84592, Bloomington), AkhR[RA]-T2A-Gal4 (#Gr046, Shu Kondo), DH31R-T2A-Gal4 (#Gr050, Shu Kondo), DH44R2-T2A-Gal4 (#Gr049, Shu Kondo), PdfR[RA]-T2A-Gal4 (#Gr090, Shu Kondo), Gyc76C-T2A-Gal4 (Shu Kondo), UASJFRC81GFP, LexAop-transtimer/cyo;10xUASCaLexA/TM2 (#86325 / T570, Bloomington + Barnea lab) strains and crosses were maintained at 25°C.

dotGal4UASGFP/cyo;tubgal80^{ts} (#67608, Bloomington), UAS-PDFR-RNAi, W¹¹¹⁸ (5905) (Barnea lab), UAS-Gyc76C-RNAi, UAS-AKHR-RNAi (TRIP) [(#29577) TRIP/Bloomington], UAS-DH31R-RNAi (#25925, Bloomington), UAS-DH44R2-RNAi (#67862, Bloomington), UAS-hid,rpr (Wegener lab stock) strains and crosses were raised at 25°C. As soon as after the eclosion, flies were shifted at 29°C.

Receptor Gal4 lines were crossed with UASJFRC81GFP reporter line to perform expression analysis and validate the single-cell data. For loss of function analysis, UAS-RNAi lines were crossed with nephrocyte-specific driver dotGAL4UASGRP/cyo; tubgal80^{ts} (#), to have knock down in nephrocytes.

LexAop-transtimer/cyo;10xUASCaLexA/TM2 line was crossed with PdfR[RA]-T2A-Gal4 to be able to measure Ca²⁺ variation in pericardial nephrocytes.

The experimental flies were grown in an uncrowded condition with a regular photoperiod (12 h light: 12 h dark).

Females were separated from males and placed on standard agar food the day before dissection to allow egg release and to maintain consistent caloric intake, as the amount of yeast in the vials is not always constant.

Fixation Flies

Prepare fixative solution with 1.5 mL 4% paraformaldehyde in Phosphate Buffered Saline (4% PFA in PBS, ThermoScientific J19943-K2) + 200 µL 0.5% Triton-X100 in PBS (PBS-T) under fume hood and placed in 2 ml tube. Anesthetize adult 7-day-old female flies of the genotype under investigation with CO₂ and put in the tube with the fixation solution. Place the tube on the nutator at room temperature (RT) for 4 hours. After fixation time, remove fixation solution and wash flies in 0.5% PBS-T on the nutator for 4 x 15 min at RT. Remove the last wash and store the samples in PBS at 4°C. The flies are ready for dissection.

Pericardial Nephrocytes Dissection

Dissect 10 cuticles of genotype under investigation, with hearts and pericardial nephrocytes, in cold PBS on a dissection pad. Put the flies on the dorsal side and remove the ventral cuticle with the scissors. Gently, remove gut and eggs and store in a 0.5 mL LoBind tube [Eppendorf Protein LoBind Tubes 0.5 mL, PCR clean, CS/100 (2x 50/PK)] containing the blocking solution placed on ice.

Blocking

To permeabilize and block unspecific antibody binding, use blocking solution containing 5% normal goat serum (NGS) in PBS-T + Sodium azide (dilution 1:100). Place the tubes containing samples on the nutator and incubate for 1 hour (can go up to 2 hours) at RT. After incubation time, remove blocking solution and add the required primary antibody.

Fluorescent immunohistochemistry

Primary Antibody Solution

Add with 100 µL of primary antibody direct against recombinant full-length protein corresponding to GFP [chicken anti-GFP, Abcam (used final concentration 1:1000)] in blocking solution. Incubate at 4°C for 2 overnights. After incubation time, remove and store primary antibody solution. Wash samples for 4 x15 min in 0,5% PBS-T at RT, on the nutator and add corresponding secondary antibody.

Secondary Antibody Solution

Add 100 µL secondary antibody [AlexaFluor488, ThermoScientific (goat anti-chicken, used final concentration 1:1000), Phal-643, Invitrogen TM (phalloidin, used final concentration 1:1000), 4',6-diamidino-2-phenylindole (DAPI), Invitrogen TM (used final concentration 1:1000)] in blocking solution and incubate at 4°C for 2 overnights. After incubation time, remove and discard secondary antibody solution. Wash samples in 0.5% PBS-T on the nutator for 4 x 15 min at RT. Do a final wash with PBS for 15 min.

Calcium Imaging

Anesthetize adult live female flies 7/8-day-old LexAop-transtimer/cyo;10xUASCaLexA/TM2>Pdf[RA]-T2A-Gal4 on ice and dissect cuticles, with hearts and pericardial nephrocytes, in ice-cold hemolymph-like solution (HLS) on a dissection pad. Put the flies on the dorsal side and remove the ventral cuticle with the scissors. Gently, remove gut and eggs and store in a 0.5 mL LoBind tube with HLS.

Add 100 μ M of serotonin or octopamine (Stock 10 mM) in 400 μ L of HLS and incubate for 15 minutes, protected by light. Then, wash one time with HLS for a minute, in order to remove the excess of the ligand. Fix the cuticles with 4% PFA/PBS + 0.5% PBS-T, for 1 hour on nutator at RT. After fixation time, wash 0.5% PBS-T on the nutator for 4 x 15 min at RT. Add blocking solution and place the tubes on the nutator; incubate for 1 hour at RT.

After incubation time, remove blocking solution and perform immunohistostaining. Add with 400 μ L of primary antibody direct against recombinant full-length protein corresponding to GFP [chicken anti-GFP, Abcam (used final concentration 1:1000)] in blocking solution. Incubate on shaker at 4°C for one overnight. After incubation time, remove and store primary antibody solution. Wash samples for 4 x 15 min in 0,5% PBS-T at RT, on the nutator. Add 400 μ L secondary antibody [AlexaFluor488, ThermoScientific (goat anti-chicken, used final concentration 1:2000)] in blocking solution and incubate on shaker at 4°C for one overnight. After incubation time, remove secondary antibody solution and wash samples in 0.5% PBS-T on the nutator for 4 x 15 min at RT. Do a final wash with PBS for 15 min. The samples are ready for mounting.

Mounting by Bridging a Cover Slip

Put two coverslips (Fisher Scientific Cat# 12-541, square. 18mm, no. 1.5) on a slide and align them to leave enough space in between where samples will be mounted. Fix the coverslips to the slide by applying glycerol-gelatine (Kaiser's glycerol gelatine phenol-free, ROTH) to the upper corners. Cut off the end of a P-200 pipet tip for transfer the cuticles onto the slide. Avoid transferring excess PBS. Use forceps and orient the dorsal cuticle side faces down, so that the heart and pericardial cells are easily accessible for imaging. Remove excess PBS, then put a coverslip and form a bridge using the 2 fixed coverslip to cover the sample. Be careful do not samples dry out. Slowly pipet 70 μ L mounting medium (Fluoromount-GTM, invitrogen by Thermo Fisher) to the topside of the coverslip. Wait mounting medium diffuse under the coverslip. Then pipet a second to the top and bottom of the cover slip. Let spread all night. The day after seal all the edges with Fixogum (Marabu).

Confocal Imaging

Third segment of each cuticle were scanned with a Leica SPE confocal microscope (Leica Microsystems) with a photomultiplier tube, pinhole size 1 airy unit and solid-state lasers (405, 488, and 647 nm) for excitation. First select a sample with transmission light and a 10-fold objective (HC APO L 10x/0,30 W U-V-I), then put a drop of oil (Type F Immersion fluid, Leica Microsystems 13614800) on a slide for 40-fold oil immersion objective (HC PL APO 40x/1,30

OIL CS2) used for the scans. The confocal stacks had a 1- μ m z-step size, 1024 $\times$ 1024 pixels and zoom factor 1. The images were captured using the optimal pixel sizes provided by the software (for good resolution and no background). 3 cuticles were scanned for each genotype for expression analysis and 20-25 samples for calcium imaging. Reconstruction of images was performed using Leica LAS AF Lite 4.0 software.

Image Processing

Fiji was used for quantification of Fluorescence Intensity and to process the images, only brightness and contrast adjustments were made to the images.

Water Content Measurements

Female 5/6-days old of the genotype under investigation were frozen overnight at -80°C and then were weighed using METTLER TOLEDO (EL204-IC) analytical balance. Fresh weight was determined, then flies were desiccated overnight at 60°C in the oven and weighed again. The amount of water was calculated using *Formula 1*:

$$\text{Water content} = \text{fresh weight} - \text{dry weight}$$

At least 3 replicates, each consisting of 25 flies, were tested per genotype.

Defecation Assay

Prepare blue food with 2% Agarose (ROTH), 2% Extract yeast (ROTH) and 4% Sucrose (ROTH) in 100 mL of distilled water. Heat in microwave until completely dissolved. Mix it with 1 mL of 0.5% Brilliant Blue (Ruth). Pour liquid food in a container and then put Corning Pyrex glass tubes (5 or 7mm diameter by 65mm length) so that these can be partially filled with blue food. Then put at 4°C until solidification.

Anesthetize flies of the genotype under investigation with CO₂, put 10 adult female flies in 10 separate tubes with the blue food and let them feed for 36 h at 29°C. Then, flip them individually in 10 blank tubes. Measure excretion rate as the number of blue dots (number of defecation event) per fly every 2, 4 and 6 hours. 3 replicates per genotype were performed.

Data Analysis

Analyze data with Prism. Two tailed t-test was performed for evaluation of significant statically of Fluorescence Intensity measurement. One-way ANOVA was used to analyzed data of

defecation assay and water content measurement. P values are indicated by asterisk symbols (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$). Data are expressed as means $\pm$ standard error.

Connectome Analysis

neuPRINT+ (<https://neuprint.janelia.org/?dataset=manc:v1.0&qt=findneurons>) was used for looking the putative neurohormonal neurons in VNC.

NeuroNLP.MANC (<https://manc.neuronlp.fruitflybrain.org/>) and *neuPRINT+* were used to obtain images of putative neurohormonal neurons and VNC.

RESULTS

Anatomical screening of neurohormone receptors

Single-cell transcriptome analyses of nephrocytes were used to screen the receptors present in pericardial cells. An expression analysis using the Gal4/UAS system was employed, to validate the gene expression. In this approach, two distinct *Drosophila* parental strains are required. One strain carries the gene encoding the Gal4 transcription factor fused with the gene of the receptors under investigation. The second strain contains the green fluorescent protein (GFP) reporter gene controlled by the Upstream Activator Sequence (UAS), to which Gal4 specifically binds, activating gene transcription. Various crosses were performed to obtain progeny with both system components (Brand and Perrimon, 1993). If the receptor is expressed, GFP is produced in the pericardial nephrocytes.

Alpha2-adrenergic-like octopamine receptor (Oct α 2R)

The Alpha2-adrenergic-like octopamine receptor (Oct α 2R) is a GPCR coupled with an inhibitory G-protein of the rhodopsin-like class, encoded by the *CG18208* gene. This receptor is predominantly identified in the heads, thoraxes, and legs of adult flies. Octopamine, tyramine, epinephrine, and norepinephrine induce a dose-dependent reduction of cAMP synthesis due to the activation of the G-protein coupled receptor, which subsequently inhibits the activity of adenylate cyclase (Qi *et al.*, 2017). Oct α 2R signaling plays a role in modulating locomotion, grooming, and hyperactivity induced by starvation (Nakagawa *et al.*, 2022).

The cross of Oct α 2R-T2A-Gal4 > UAS-JFRC81GFP did not exhibit GFP gene reporter expression in the nuclei of pericardial nephrocytes, which are the large nuclei located along each side of the heart tube (as shown in Figure 9). This lack of expression indicates the absence of Oct α 2R expression in these cells, but other isoforms must be present.

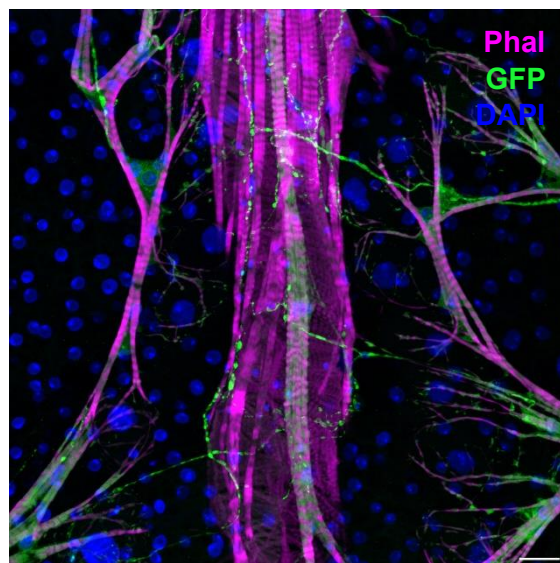


Figure9: No expression of GFP in Oct-alpha-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, large nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μ m.

5-hydroxytryptamine (serotonin) receptor 7 (5-HT7)

5-hydroxytryptamine (serotonin) receptor 7 (5-HT7) is a protein of 564 amino acids, which possesses eight putative transmembrane domains and exhibits significant homology with all G-protein-coupled receptors. This receptor is encoded by the *CG12073* gene. When exposed to serotonin, 5-HT7 activates adenylate cyclase, leading to a concentration-dependent and saturable increase in cAMP levels (Witz *et al.*, 1990). In the third larval instar brain, 5-HT7 is expressed and localized within specific circuits in the brain hemispheres, as well as in neurons within the ventral ganglion. In the adult brain, it is found in large-field R neurons that innervate the ellipsoid body, as well as in neurons closely associated with PDF-positive ventrolateral neurons clock neurons, which innervate the optic lobes. Moderate expression of 5-HT7 is detected in the olfactory and gustatory regions of the brain, with weaker expression observed in other central complex structures such as the fan-shaped body. Notably, 5-HT7 expression appears to be post-synaptic in both larvae and adults (Becnel *et al.*, 2011). Furthermore, 5-HT7 is expressed in various other contexts, including the *Drosophila* mushroom body (Ganguly *et al.*, 2020), prothoracic gland cells (Shimada-Niwa and Niwa, 2014), proximal-targeting flexor and extensor motor neurons, proprioceptive neurons, and distal sensory neuron populations in the VNC (Howard *et al.*, 2019). Abundant expression of 5-HT7 is observed in the proventricular region, as well as in processes that innervate the proventriculus/anterior midgut and the crop (Yao and Scott, 2022).

5-HT7 has various functions, including a role in courtship and mating behavior in flies (Becnel *et al.*, 2011) and the regulation of virgin female receptivity (Ma *et al.*, 2022). It also plays a crucial role in olfactory learning and memory processes, encompassing acquisition, consolidation, and retrieval (Johnson *et al.*, 2011). In larvae, it acts as an important modulator of motor programs (Silva *et al.*, 2014) and is essential for appetitive olfactory learning (Ganguly *et al.*, 2020). Additionally, 5-HT7 influences feeding microstructure (Banu *et al.*, 2023), regulates behaviors dependent on feeding state (Vogt *et al.*, 2021), promotes crop contractions in the presence of bitter and potentially harmful substances (Yao and Scott, 2022), regulates sleep and sleep structure (Liu *et al.*, 2019), plays a pivotal role in ecdysteroid biosynthesis in the prothoracic gland (Shimada-Niwa and Niwa, 2014), and influences walking speed (Howard *et al.*, 2019). In the VNC, it plays a role in regulating fly immobility (Gowda *et al.*, 2022).

As shown in *Figure 10*, the cross between the 5-HT7-T2A-Gal4 and UASJFRC81GFP strains reveals the expression of the GFP gene reporter within the nuclei of pericardial nephrocytes, identified by their large nuclei positioned on either side of the heart tube, then this observation suggests the presence of 5-HT7 expression within these specific cells.

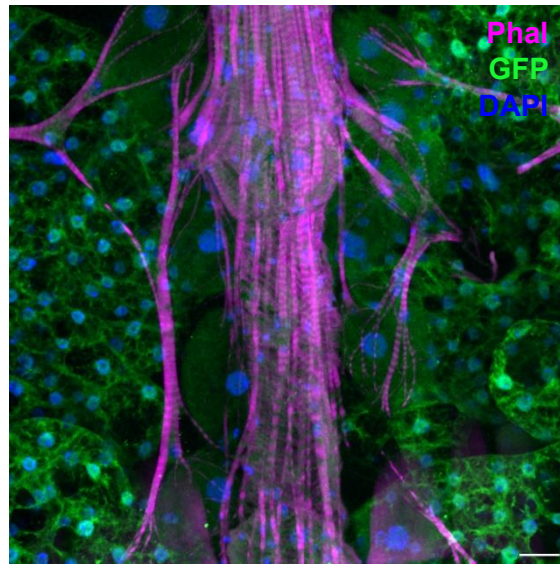


Figure 10: Expression of GFP in 5-HT7-T2A-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, large nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μm .

Adipokinetic hormone receptor (AkhR)

The Adipokinetic hormone receptor (AkhR), encoded by the *CG11325* gene, exhibits structural and evolutionary similarity to vertebrate gonadotropin-releasing hormone receptors. This receptor consists of 831 amino acid residues and features 7 membrane-spanning regions, characteristic of G-protein-coupled receptors (GPCRs) (Hauser et al., 1997). Upon activation of the G_q protein, a signaling cascade is initiated that necessitates an upsurge in inositol (1,4,5) triphosphate (IP3). This increase promotes phospholipase C (PLC) stimulation, culminating in the synthesis of IP3 and DAG. Subsequently, binding of IP3 to specific receptors within the endoplasmic reticulum triggers the release of Ca^{2+} , which in turn activates glycogen phosphorylase. Simultaneously, DAG activates protein kinase C (PKC), leading to the activation of phosphorylase α and the formation of glucose-1-phosphate (Gäde and Auerswald, 2003). AkhR plays a pivotal role in the accumulation and mobilization of stored fat, contributing to both lipid and carbohydrate catabolism. AkhR was detected in the fat body of *Drosophila*, as well as in a

subset of neurons within the adult subesophageal ganglion, neurons associated with the perception of attractive tastes (Bharucha *et al.*, 2008).

As shown in *Figure 11* in this model, the cross between AKHR-T2A-Gal4 and UASJFRC81GFP shows the expression of a GFP gene reporter within the nuclei of pericardial nephrocytes (large nuclei flanking the heart tube).

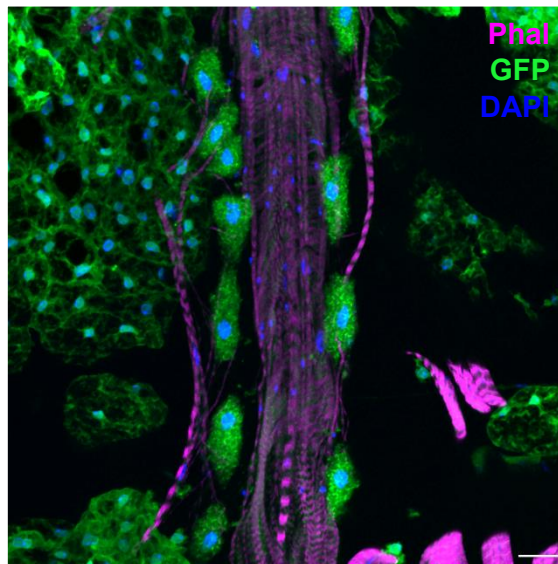


Figure 11: Expression of GFP in AKHR-T2A-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, large nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μm .

Diuretic hormone 44 receptor 2 (Dh44-R2)

Diuretic hormone 44 receptor 2 (DH44R2) is a Class B or secretin GPCR member of the corticotropin-releasing factor (CRF) receptor subfamily, encoded by the *CG12370* gene. An identifying characteristic of Class B GPCRs is the presence of vast extracellular domains (Pal *et al.*, 2012). After applying DH44 peptide, this receptor become activated, leading to the specific translocation of β -arrestin to the PM and the initiation of cAMP signaling pathways in expressing cells. DH44R2 expression is observed in Malpighian tubule principal cells (Hector *et al.*, 2009) and enteroendocrine cells within the gut (Dus *et al.*, 2015; Guo *et al.*, 2019). Its functional roles encompass the homeostatic regulation of osmotic balance (Hector *et al.*, 2009), facilitation of proper intestinal motility and excretion (Dus *et al.*, 2015), contribution to desiccation and starvation tolerance (Cannell *et al.*, 2016), and control of amino acid consumption (Yang *et al.*, 2018).

The DH44R2-T2A-Gal4 > UASJFRC81GFP cross demonstrates GFP gene reporter expression within the nuclei of pericardial nephrocytes, which are characterized by their large

nuclei located along each side of the heart tube, as depicted in *Figure 12*, suggesting the presence of DH44R2 expression in these cells.



Figure 6: Expression of GFP in DH44R2-T2A-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, big nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μm .

Diuretic hormone 31 receptor (Dh31R)

The Diuretic hormone 31 Receptor (DH31R) belongs to the Class B1 GPCR category and shares relatedness with members of the calcitonin receptor-like receptor (CLR) family, encoded by the CG32843 gene. Class B1 GPCRs carry out their signaling through a stimulatory G-protein, which in turn governs intracellular cAMP levels by associating with adenylate cyclase (Harmar, 2001). It has been demonstrated that DH31R, in conjunction with the receptor component protein (dRCP), responds to DH31 signaling. DH31R is expressed in various cellular contexts, including the principal cells of both larval and adult Malpighian tubules, corazonin-expressing cells (Johnson *et al.*, 2005), pacemaker dorsal neurons DN1s and DN3s, later neurons 1-LNVs and short neuropeptide F-expressing cells (Goda *et al.*, 2018), the anterior region of the visceral muscle (Benguettat *et al.*, 2018), as well as in neurons expressing Allatostatin A, Leucokinin, Drosulfakinin, and Allatostatin C (Lin *et al.*, 2022). DH31R plays multiple roles, including modulating bodily fluid secretion (Johnson *et al.*, 2005), contributing to the circadian regulation of sleep mechanisms (Kunst *et al.*, 2014), governing the daytime temperature preference rhythm (Goda *et al.*, 2018), regulating visceral muscle contractions (Benguettat *et al.*, 2018), influencing

dietary amino acid aphrodisiac effects, suppressing protein feeding, and facilitating the transition from feeding to courting behaviors (Lin *et al.*, 2022).

The cross between DH31R-T2A-Gal4 and UASJFRC81GFP resulted in the expression of a GFP gene reporter, which was observed within the nuclei of pericardial nephrocytes (large nuclei situated on both sides of the heart tube, as illustrated in *Figure13*). This expression pattern of GFP provides evidence supporting the presence of DH31R expression within these specific cells.

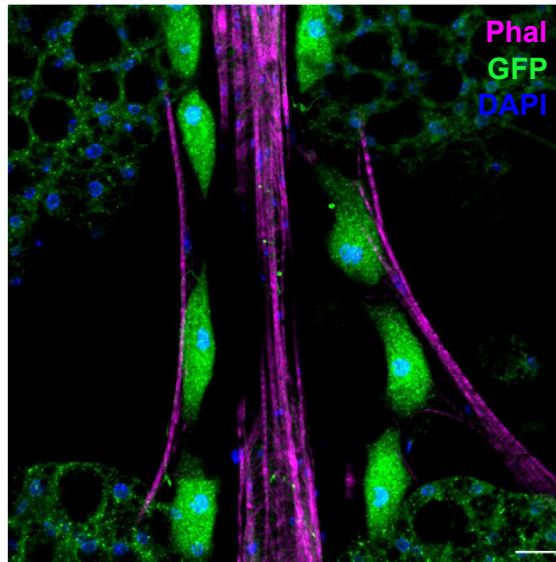


Figure13: Expression of GFP in DH31R-T2A-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, large nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μm .

Pigment-dispersing factor receptor (PdfR)

The Pigment-dispersing factor receptor (PdfR) belongs to Class II GPCRs, specifically the secretin-related class, encoded by the *CG13758* gene. It is known to elevate cAMP levels upon activation by PDF. Class II GPCR signaling shares close similarities with the calcitonin-CGRP receptor family (Mertens *et al.*, 2005), PdfR expression is prominent in the adult brain, where it is found in 13 neurons within each hemisphere. These neurons encompass the four large ventrolateral neurons (l-LNV), one dorsolateral neuron (LNd), seven dorsal neurons of group 1 (DN1), and one DN3 neuron (Hyun *et al.*, 2005). Additionally, PdfR is expressed in sub-retinal glial cells at the retina-lamina boundary (Im and Taghert, 2010), as well as in glial cells within the distal medulla neuropil (Goda *et al.*, 2018). Furthermore, it is found in the ureter muscles of the excretory system (Talsma *et al.*, 2012) and is believed to play a role in regulating AKH release, the primary hormone responsible for the mobilization of stored energy (Braco *et al.*, 2022). PdfR contributes to the regulation of circadian rhythms (Hyun *et al.*, 2005; Mertens *et al.*, 2005), impacting morning

behavior, evening phase, and circadian period (Lear *et al.*, 2009), and helps shift the balance of daily circadian activity from evening to morning (Choi *et al.*, 2012). Additionally, PdfR plays a role in the myotropic effects of ureter muscles (Talsma *et al.*, 2012), serves as a regulator of flight circuit development and is a modulator of acute flight (Agrawal *et al.*, 2013). Finally, PdfR is involved in regulating the preferred temperature decrease at night-onset (Goda *et al.*, 2017).

Our model resulting by the cross between PdfR-T2A-Gal4 and UASJFRC81GFP reveals in the nuclei of pericardial nephrocytes, distinguishable by their large nuclei positioned on either side of the heart tube the expression of the GFP gene reporter, as illustrated in *Figure 14*.

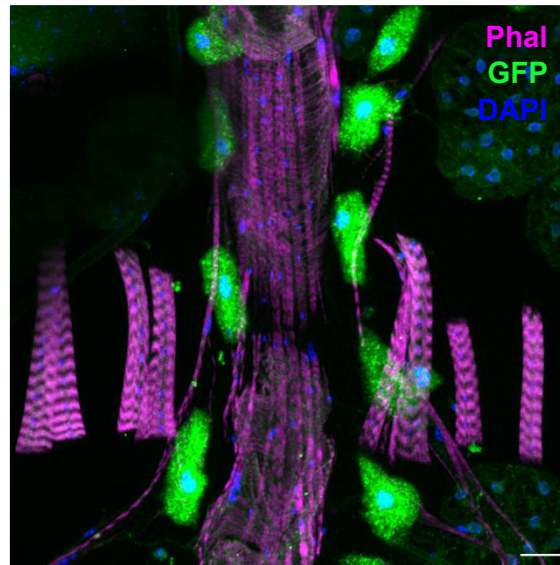


Figure 14: Expression of GFP in PdfR-T2A-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, large nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μ m.

Guanylyl cyclase at 76C (Gyc76C)

Guanylyl cyclase at 76C (Gyc76C) is a homodimeric membrane guanylyl cyclase receptor characterized by an extracellular ligand-binding domain, a short transmembrane region, and an intracellular domain with the catalytic region at the C-terminus, that catalyzes the synthesis of cGMP upon binding with a ligand (Kuhn, 2009). It is encoded by *CG42636* gene. Analysis of gene expression in *Drosophila* tissues reveals that Gyc76C is notably abundant in epithelial tissues during both larval and adult stages. These epithelial tissues include the gut, MTs, and ovaries (flyatlas2.org, Krause *et al.*, 2022). Gyc76C plays multifaceted roles, including in somatic muscle development (Patel *et al.*, 2012), modulation of organismal responses to salt stress (Overend *et al.*, 2012), and in invagination, collective migration, and lumen shape control of the *Drosophila* embryonic salivary gland (Patel and Myat, 2013). It is also required in motoneurons for axon

guidance (Chak and Kolodkin, 2014), and for *de novo* lumen formation in the *Drosophila* embryonic trachea (Myat and Patel, 2016). Additionally, Gyc76C is an immune receptor that differentially mediates both cellular and humoral immune responses (Iwashita *et al.*, 2020).

The Gyc76C-T2A-Gal4 > UASJFRC81GFP cross shows GFP expression in nuclei of pericardial nephrocytes (large nuclei located along each side of the heart tube, as show in *Figure15*), suggesting the presence of Gyc76C expression in these cells.

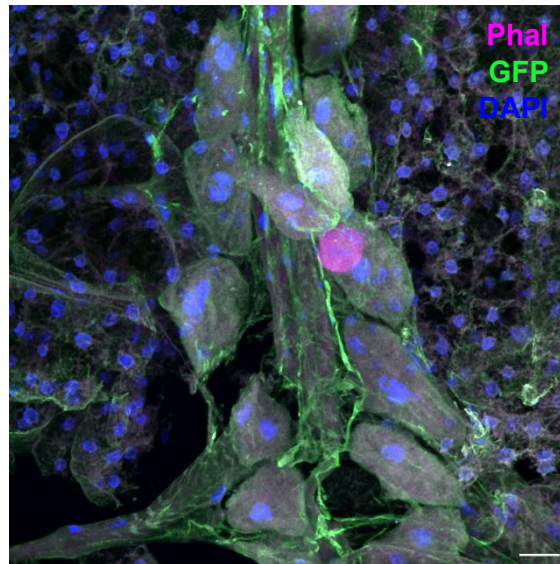


Figure 15: Expression of GFP in Gyc76C-T2A-Gal4 driven GFP expression in pericardial nephrocytes. In the middle is heart tube, large nuclei are the pericardial nephrocytes, small nuclei are the fat cells. In magenta: phalloidin (Phal), stained the cytoskeleton and heart tube; in green: GFP; in blue: DAPI, labelled the nuclei of the cells. Scale bar: 20 μm .

Calcium Imaging

To deeper understand the neurohormonal regulation of pericardial nephrocytes, calcium imaging experiments were conducted using a fluorescent calcium indicator GCaMP, that will be activate through the Ca^{2+} signaling pathway, allowing to assess if these cells respond to neurohormones. Being our pericardial nephrocytes close to the live heart, it was not easy to obtain an accurate measurement of the fluorescence intensity; thus, to address this issue, it has been opted for an alternative calcium indicator, CaLexA.

Calcium-dependent nuclear import of LexA or CaLexA for short, is based on the Nuclear Factor of Activated T Cells (NFAT) transcription factor. NFAT is transported into the cell nucleus in a manner directly influenced by cellular activity thanks its regulatory domain that contains a nuclear localization signal. NFAT's translocation into the nucleus is regulated by phosphatase Calcineurin, which is dependent on calcium and calmodulin. Increase in Ca^{2+} concentration

activates Calcineurin that dephosphorylates NFAT, which is then shuttled into the nucleus. Inside the nucleus, NFAT induces the expression of GFP reporter gene (Masuyama *et al.*, 2012). Hence, GFP is detected in cells that are activated and have increased calcium levels.

Cells were exposed to 5-HT and OA, biogenic amines known to induce a rise in Ca^{2+} concentration (Sampson *et al.*, 2020; Balfanz *et al.*, 2005; Roeder, 1999). Both these amines are known to have diverse functions in regulating both the physiology and behavior of *Drosophila* (Rosikon *et al.*, 2023).

To assess the cellular response, the dissected samples of flies carrying the CaLexA system and PdfR[RA]-T2A-Gal4 (to have a spatial expression) were maintained in hemolymph-like solution (HLS); so they were alive and incubated them with 100 μM of serotonin, which is the designated working concentration. Following fixation and immunostaining procedures, I conducted imaging of the samples using confocal microscope. This allowed to quantify the fluorescence intensity.

Analysis of fluorescence intensity in samples that had been incubated with serotonin, a reduction was observed in comparison to the control (as shown in *Figure 16C*, ** $P < 0.0074$).

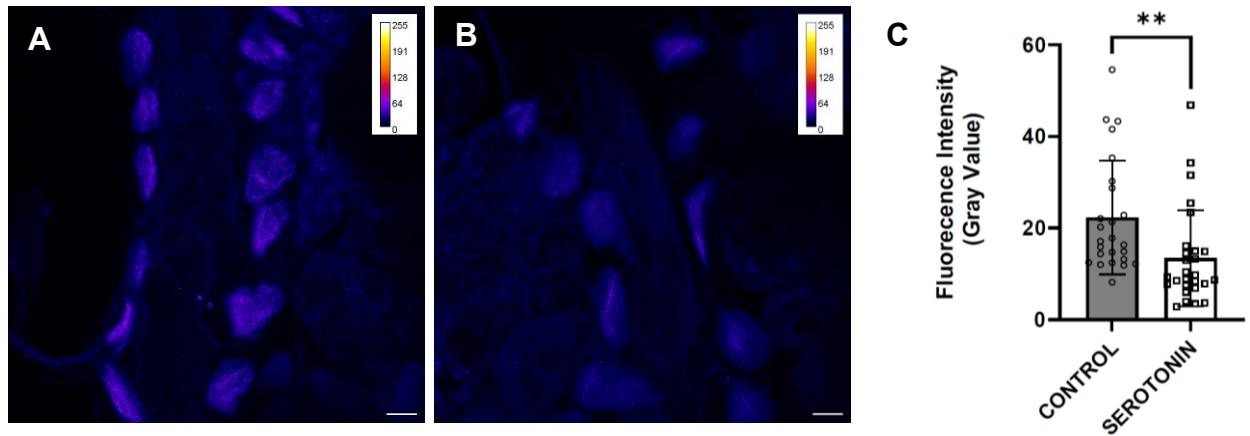


Figure 16: The fluorescence intensity of untreated (A) or treated (B) nephrocytes was quantified. C) Fluorescence intensity of pericardial nephrocytes serotonin-treated and untreated. For each condition, at least 230 nephrocytes from at least 25 adult females PdfR[RA]-T2A-GAL4 > LexAop-transtimer/cyo;10xUASCaLexA were quantified. Significance was determined by Two tailed t-test, ** $P < 0.01$. Error bars are standard error of the means. Scale bars: 20 μm .

Further, when the nephrocytes were treated with octopamine then in comparison to control samples, the nephrocytes incubated in the presence of 100 μM octopamine revealed no significant change in fluorescence intensity (as shown in *Figure 17C*, ns $P > 0.05$).

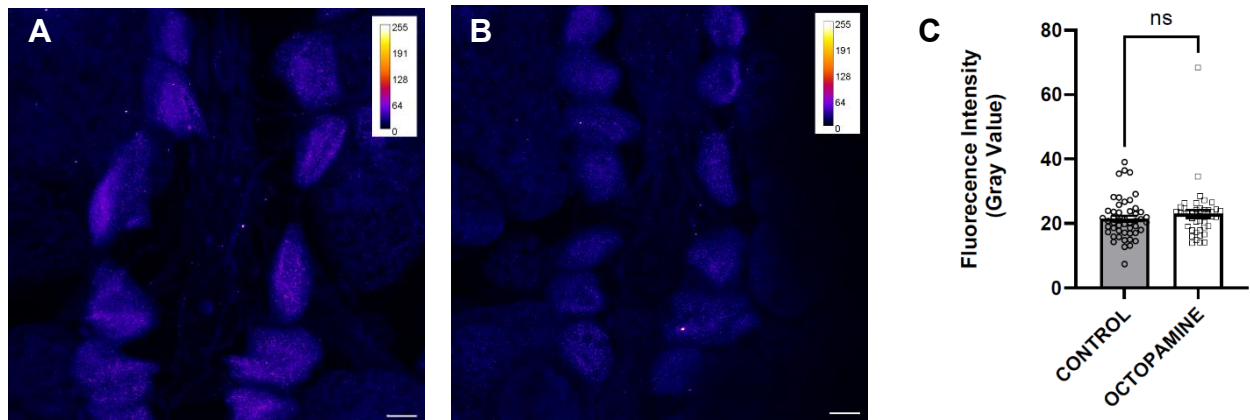


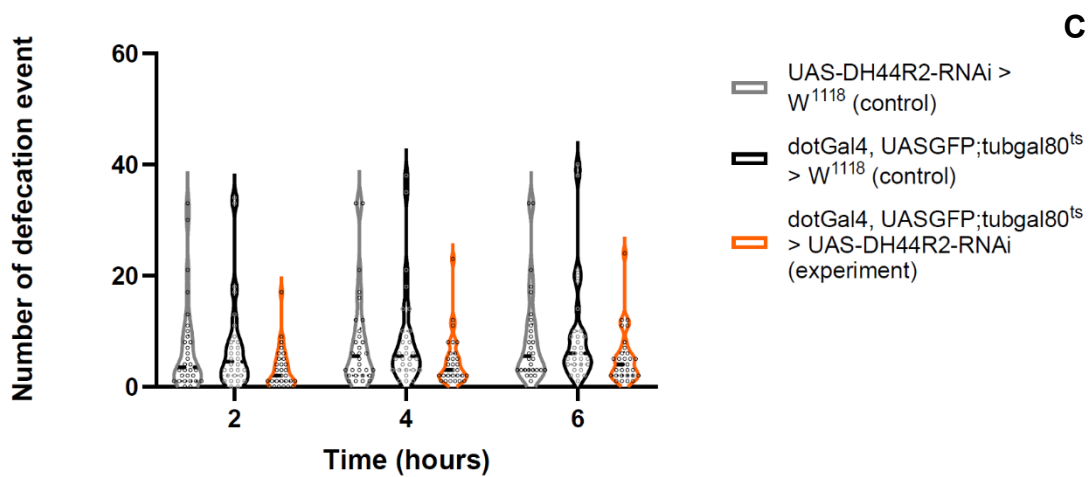
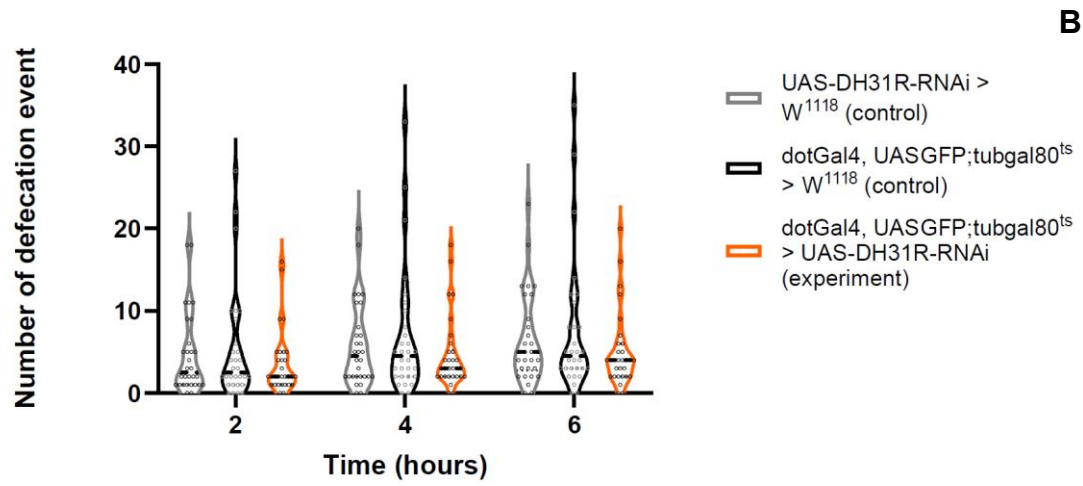
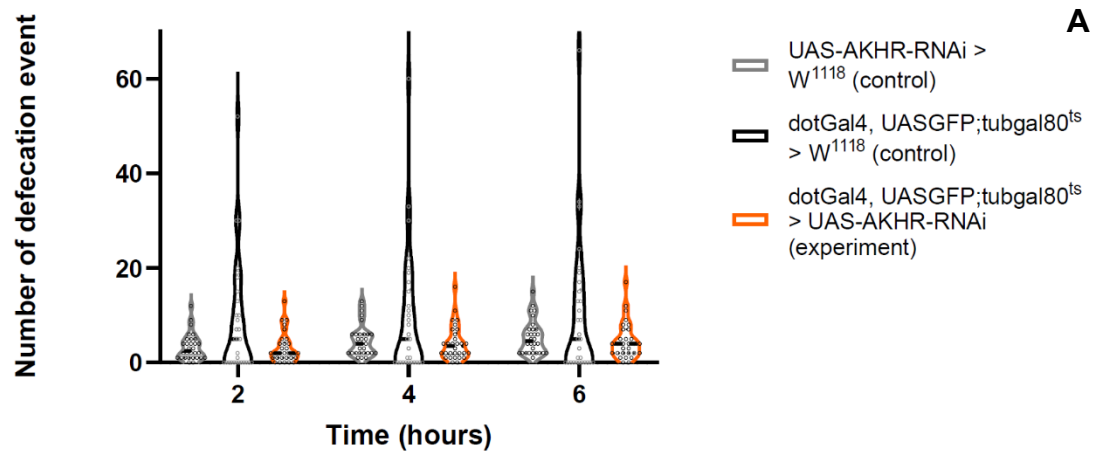
Figure 17: The fluorescence intensity of untreated (A) or treated (B) nephrocytes was quantified. C) Fluorescence intensity of pericardial nephrocytes octopamine-treated and untreated. For each condition, at least 410 nephrocytes from at least 43 adult females PdfR[RA]-T2A-GAL4 > LexAop-transtimer/cyo;10xUASCaLexA were quantified. Significance was determined by Two tailed t-test, ns $P > 0.05$. Error bars are standard error of the means. Scale bars: 20 μm .

Knock down of receptors using RNAi

After establishing the presence of all the above mentioned in nephrocytes, our next step was to assess their role in physiological process, specifically in excretion and homeostasis of body fluids.

To investigate the impact of receptor expression in nephrocytes on excretion, I employed a temperature-sensitive, nephrocyte-specific driver (dotGal4UASGFP/cyo;tubgal80^{ts}) to conduct RNAi targeting candidate receptors during screening process. These flies were initially raised at 25°C and, upon eclosion, transferred to 29°C. This temperature shifts inactivated Gal80, thereby activating Gal4, which in turn induced UAS-RNAi targeting the receptors of interest. After reaching an age of 5 to 6 days, the flies were placed on food containing blue dye at 29°C for 36 hours, and I subsequently measured the time-dependent increase in blue-dyed feces.

The frequency of defecation events over time exhibited no statistically significant change in response to the knockdown of receptors (Figure 18 from A to E), even in the case of the hid,rpr genotype, where should be a complete deletion of nephrocytes (used as a positive control) (Figure 18F).



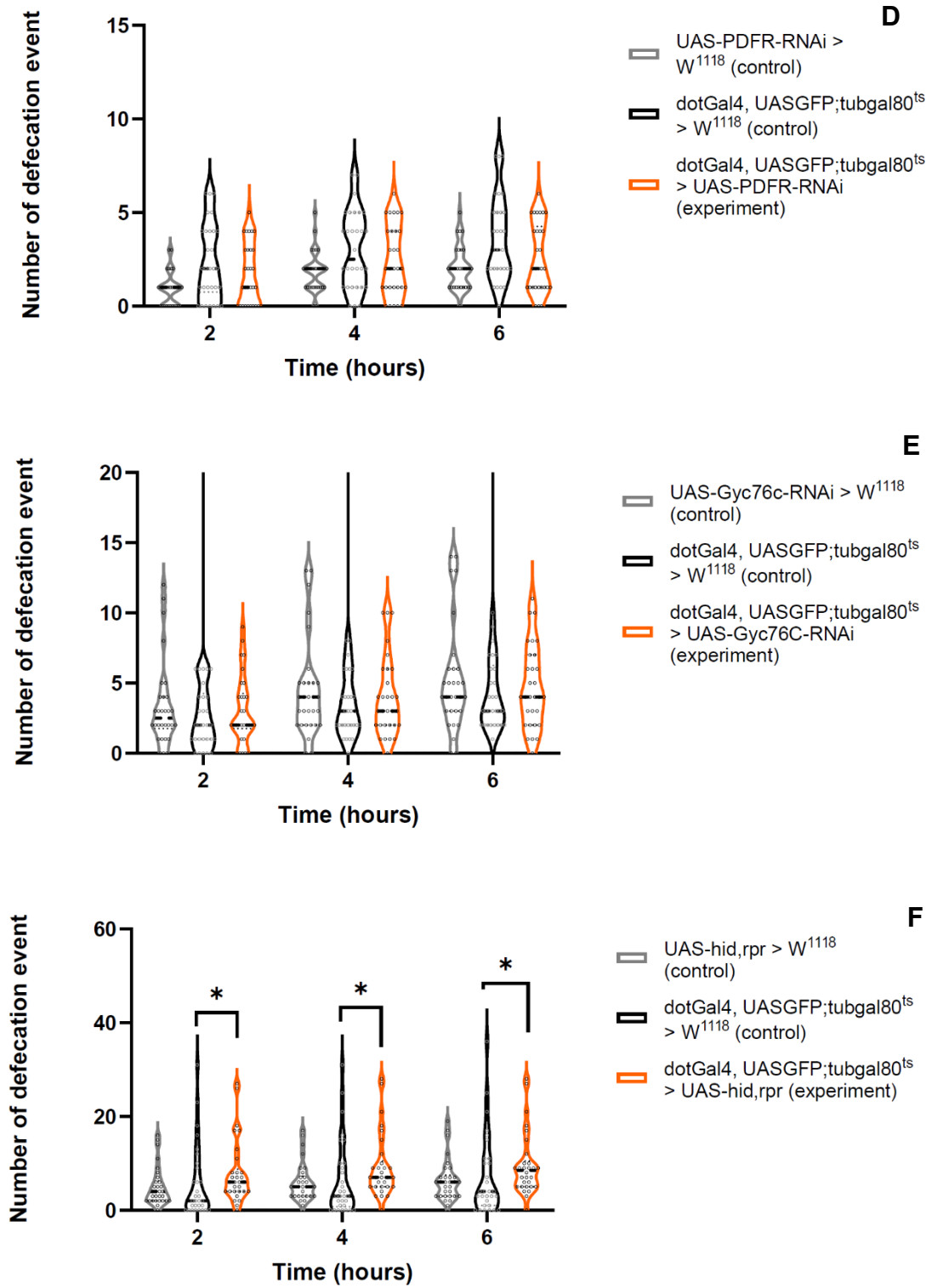
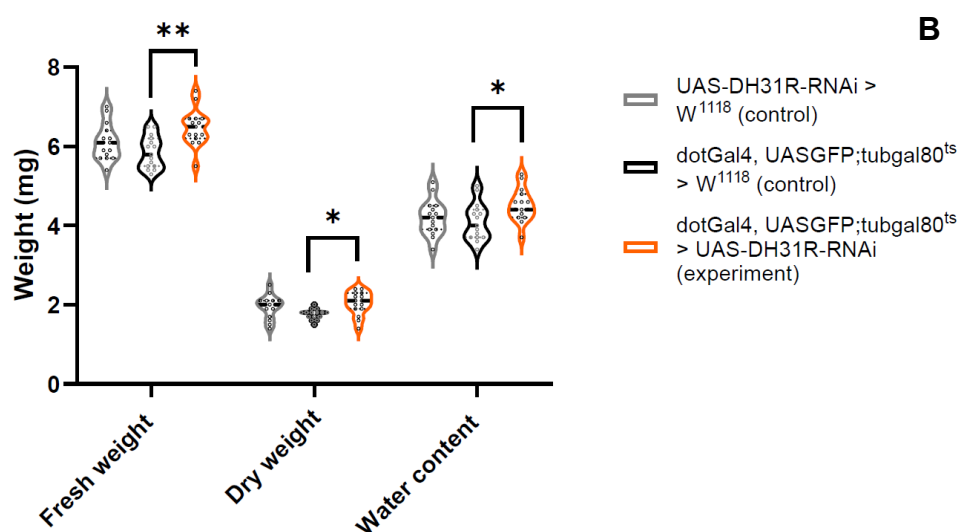
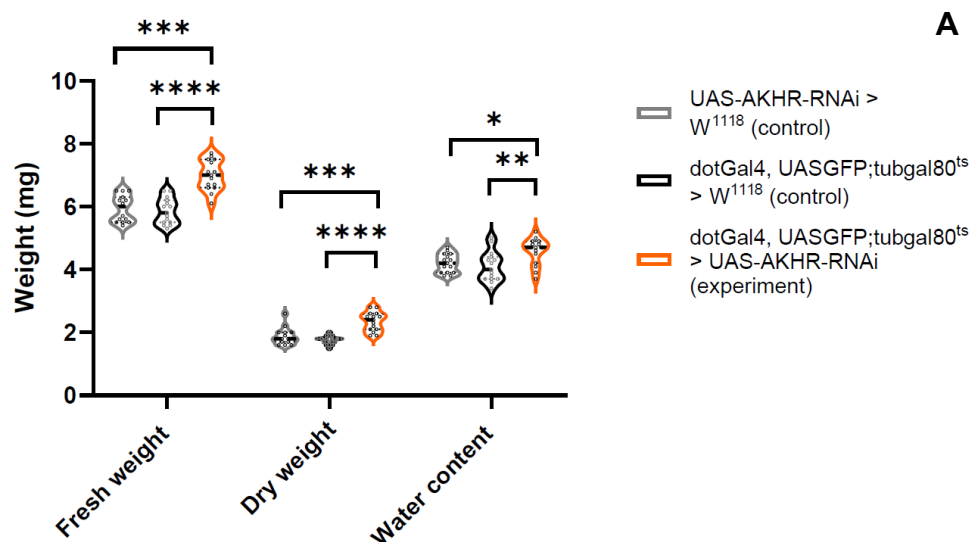
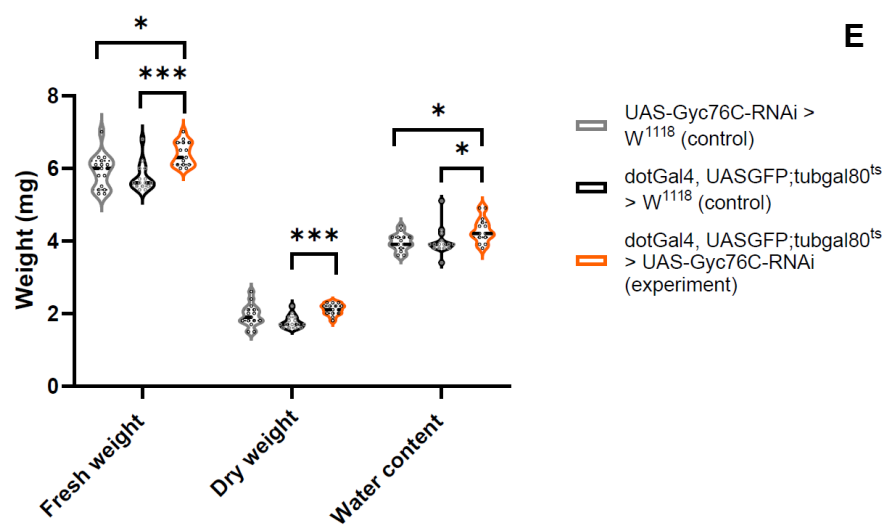
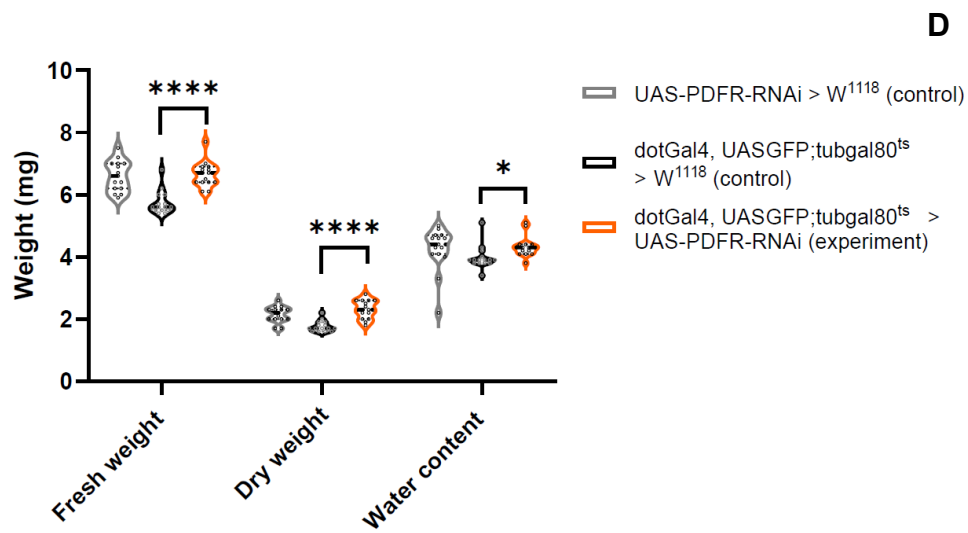
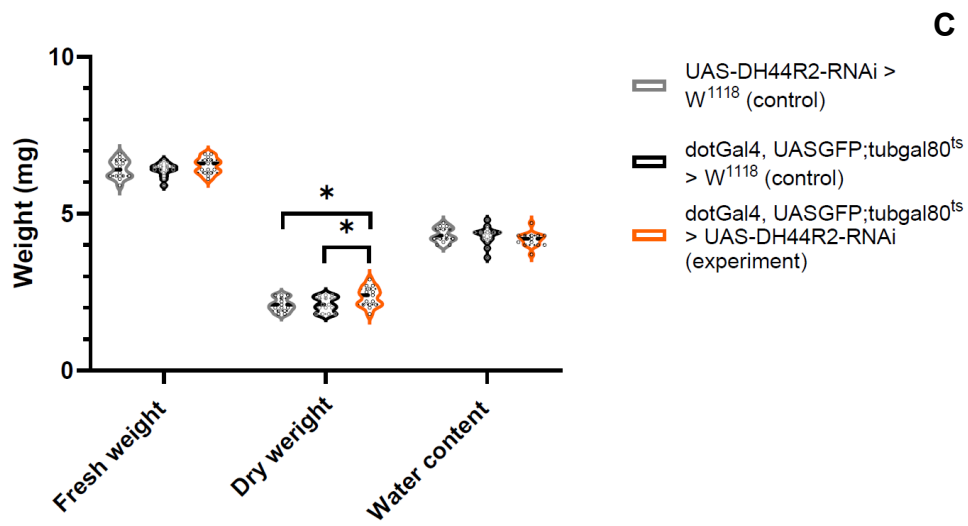


Figure 18: The knockdown of receptors that screened positive showed no observable change in defecation rate over the time. At time 2, 4 and 6 hours no change in defecation rate were found in AKHR, DH31R, DH44R2, Gyc, PDFR, and hid,rpr experiment when compared with controls. Kruskal-Wallis test: $P > 0.05$ for each comparison with controls. 3 categories of crosses: 1. First Control Crosses (grey): UAS-RNAi x w^{1118} crosses. 2. Secondary Control Crosses (black): dotGal4 x w^{1118} crosses. 3. Experimental Crosses (orange): consist of dotGal4 x UAS-RNAi crosses. UAS-hid,rpr genotype is the positive control.

To assess the potential involvement of these receptors in the maintenance of water balance, it was employed the same specific nephrocyte driver used in the defecation assay. The flies were raised under the previously mentioned conditions. At the age of 6/7 days, the flies were frozen at -80°C overnight and then weighed. Subsequently, they were placed in an oven at 60°C overnight and weighed once more. The water content was calculated using *Formula 1*.

Knock down of the AKH receptor resulted in alterations in both fresh weight and dry weight, accompanied by changes in water content (*Figure19A*). In contrast, *Gyc76C* exhibited fluctuations in fresh weight and water content but no statistically significant change in terms of dry weight (*Figure19E*), similarly shown by *hid,rpr* (*Figure19F*). Conversely, *DH44R2* displayed differences exclusively in dry weight (*Figure19C*). Knock down of *DH31*- and *PDF*-receptor did not exhibit any statistically significant differences (*Figure19B* and *D*, respectively).





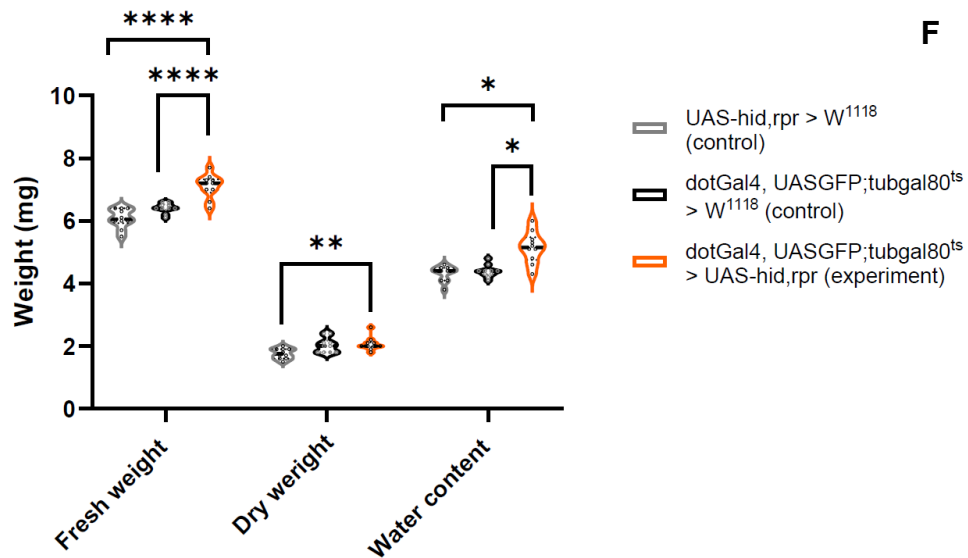


Figure 19: Statistical analysis revealed differences in fresh weight, dry weight, and water content between the receptor knockdown experiments and the control groups. Interference with the AKH receptor (A) resulted in significant changes in fresh weight and dry weight (Kruskal-Wallis test: approximate $P < 0.0001$), as well as water content (Ordinary one-way ANOVA $P < 0.0053$). On the other hand, interfering with the DH44R2 receptor (C) only led to a significant difference in dry weight (Ordinary one-way ANOVA $P < 0.0076$). Interference with Gyc76c (E) induced differences in both fresh weight (Kruskal-Wallis test: approximate $P < 0.0004$) and water content (Kruskal-Wallis test: approximate $P < 0.0060$). Similar alterations in fresh weight (Ordinary one-way ANOVA $P < 0.0001$) and water content (Kruskal-Wallis test: approximate $P < 0.0185$) were observed in hid,rpr genotype. In contrast, DH31R (B) and PDFR (D) showed no statistically significant changes in any of the three weight categories. DH44R2 did not exhibit any significant differences in fresh weight or water content. Gyc76c and hid,rpr showed no significant differences in dry weight, Kruskal-Wallis test and Ordinary one-way ANOVA ($P > 0.05$ for each comparison with controls). 3 categories of crosses: 1. First Control Crosses (grey): UAS-RNAi x w¹¹¹⁸ crosses. 2. Secondary Control Crosses (black): dotGal4 x w¹¹¹⁸ crosses. 3. Experimental Crosses (orange): consist of dotGal4 x UAS-RNAi crosses.

Identifying putative neurons which produce neurohormone in *Drosophila* VNC

In *Drosophila*, both the brain and the VNC contain cells responsible for hormone production. While research on the brain cells has been recently conducted (Reinhard *et al.*, 2023 preprint), another aim of this study was to investigate the VNC. Specifically, we focused on neurosecretory cells within the abdominal ganglia and thoracic T3 ganglia of the VNC, as these cells produce hormones Lk and DH44, that have an impact on MTs (as depicted in *Figure 20*, Zandawala *et al.*, 2018).

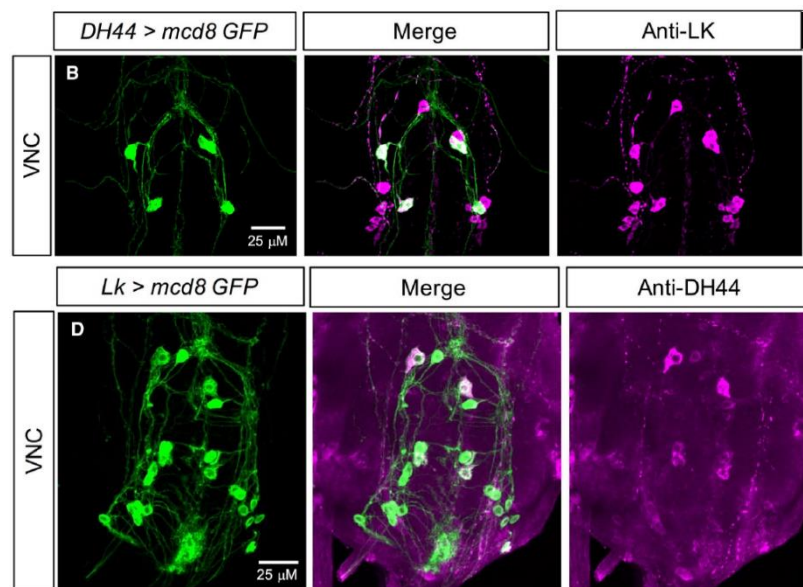


Figure 20: In adult VNC, there is colocalization observed between B) DH44-GAL4-driven GFP and LK-immunoreactivity and D) Lk-Gal4-driven GFP and DH44-immunoreactivity, both specifically within a subset of abdominal Lk neurons. *Reference images: Zandawala M, Marley R, Davies SA, Nässel DR. Characterization of a set of abdominal neuroendocrine cells that regulate stress physiology using colocalized diuretic peptides in Drosophila. Cell Mol Life Sci. 2018;75(6):1099-1115. doi:10.1007/s00018-017-2682-y (modified).*

The goal was to identify analogous neurons within the connectome, utilizing the VNC connectome as a reference (Plaza *et al.*, 2022). Neurons that produce neurohormones are notably characterized by the presence of a sizable cell body containing hormone high-dense vesicles. Building upon this distinction, it was used the *neuPRINT+* website to identify potential neurohormone-producing neurons that influence MTs. Furthermore, it was examined their shape, aiming to identifying similarities to the shapes observed via confocal microscopy, which resembled a box of a violin (*Figure20*).

Neurons such as MNad13, MNad38, MNad27, MNad18, MNad21, MNad59, MNad25, MNad52, MNad03, MNad07 and MNad37 types appear to closely resemble the morphology observed via confocal microscopy (Zandawala *et al.*, 2018), in the overall visualization (*Figure 21*).

These neurons exhibit the typical characteristics of neurohormone-producing neurons. These characteristics become evident when observing them in a 3D view.

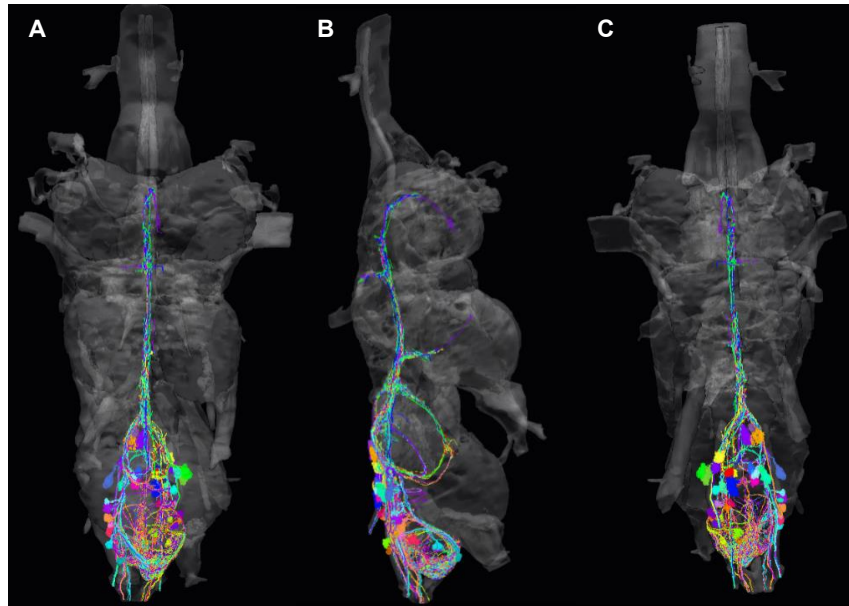
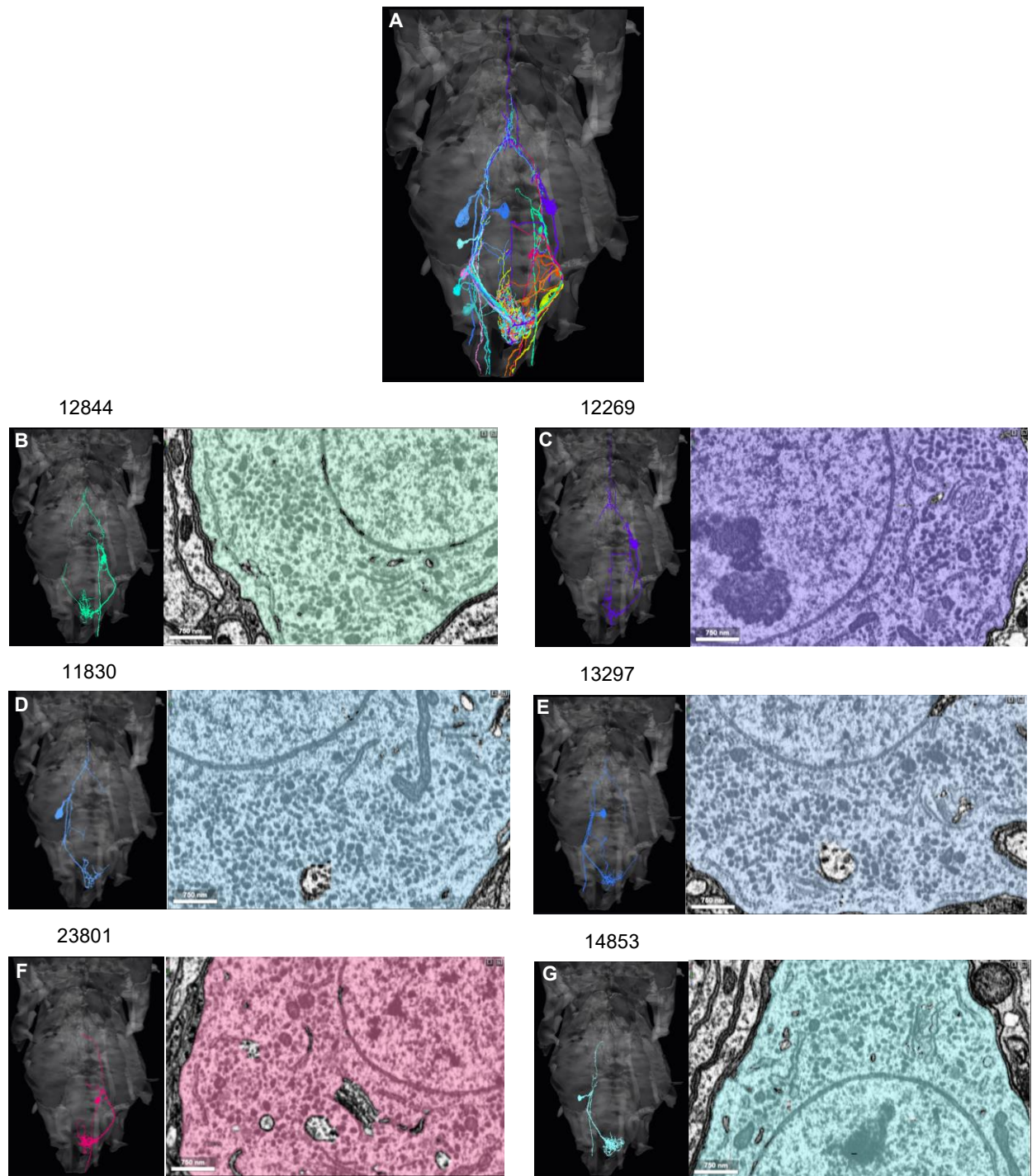


Figure 21: Comprehensive visualization of MNad neurons resembling the morphology of Lk- and DH44-producing cells in the adult female VNC. A: ventral view; B: side view; C: dorsal view.

MNad13 type comprises 12 neurons (*Figure22A*), identified by cell ID: 12844, 12269, 11830, 13297, 23801, 14853, 16178, 17576, 13655, 16375, 100612 and 15182 (*Figure22* from *B* to *M*).



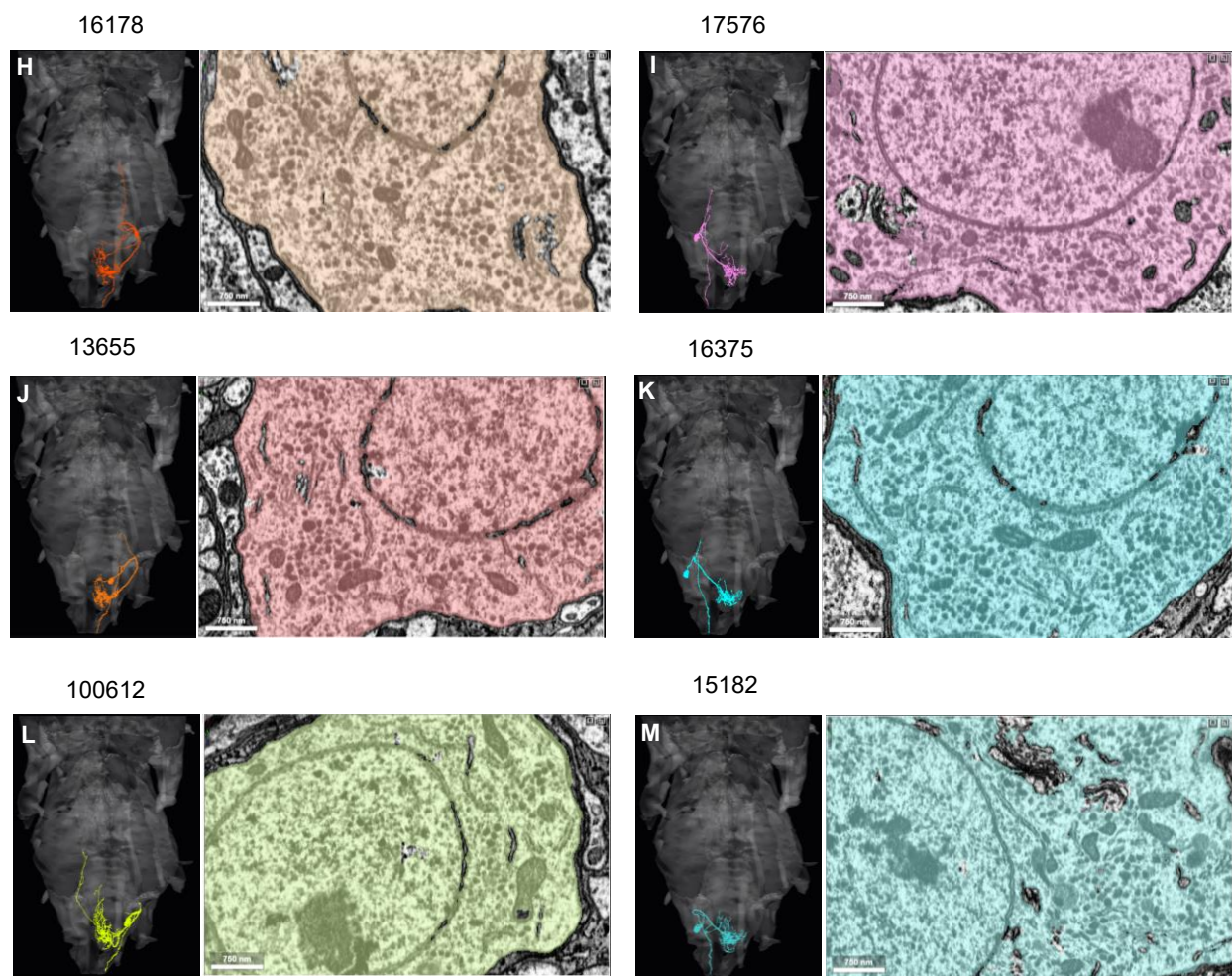


Figure 22: MNad13 type neurons. A. 3D view of neurons of MNad13. From B to M: 3D view of single neuron (on the left of each image), tangential section view at electron microscopy (on the right of each image) along with corresponding cell ID. Scale bars: 750 nm.

MNad38 type comprises a single neuron with cell ID 13050 (*Figure23*).

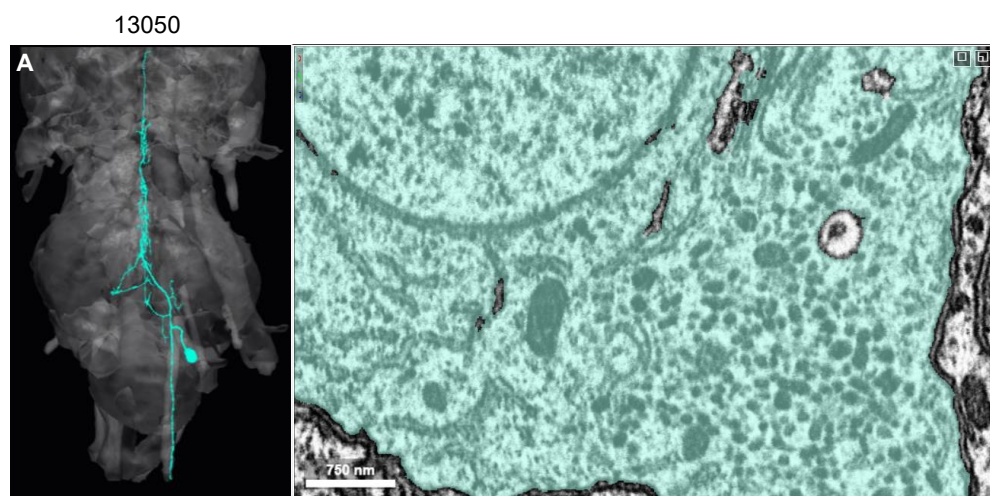


Figure 23: MNad38 type neuron. A. 3D view of single neuron (on the left), tangential section view at electron microscopy (on the right) along with corresponding cell ID. Scale bar: 750 nm.

MNad27 type consists of 2 neurons (*Figure 24A*), with cell ID 12445 and 12314 (*Figure 24 B and C*).

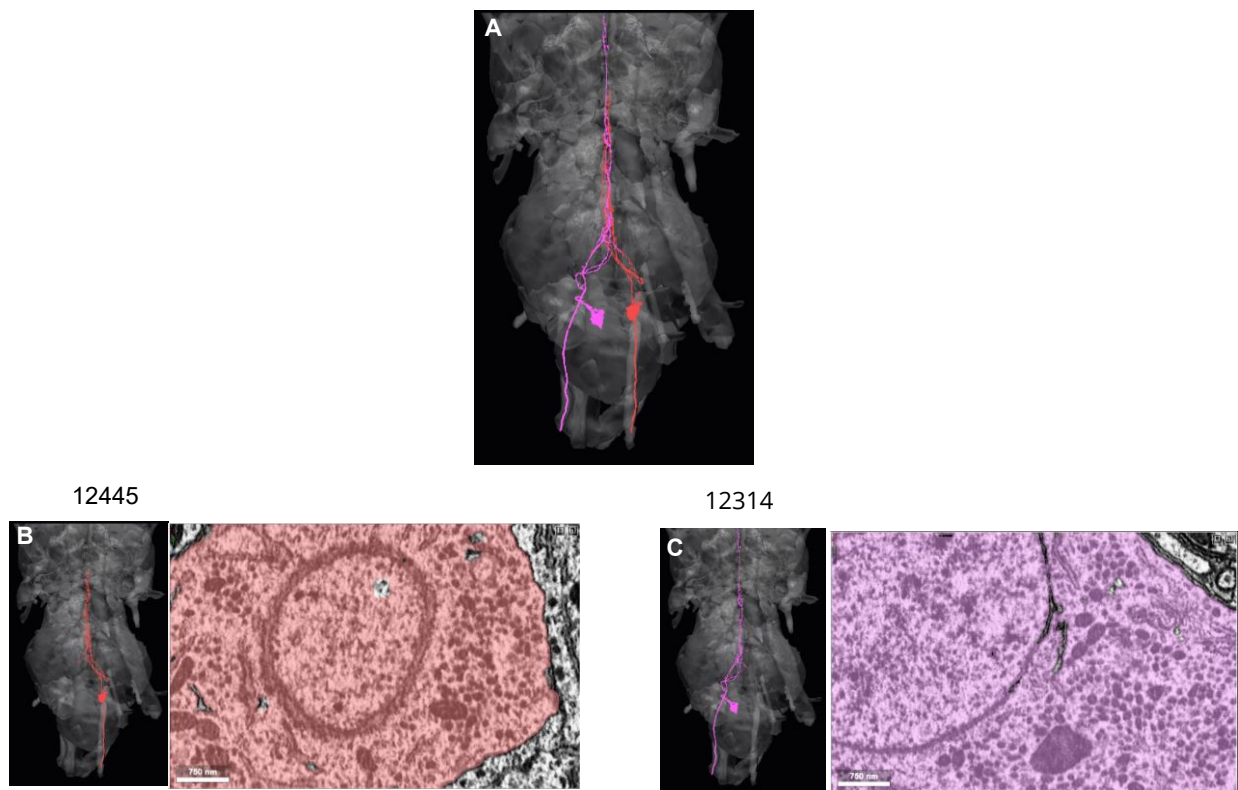


Figure 24: MNad27 type neurons. A. 3D view of neurons of MNad27. B and C: 3D view of single neuron (on the left of each image), tangential section view at electron microscopy (on the right of each image) along with corresponding cell ID. Scale bars: 750 nm.

MNad18 type consists of 4 neurons (*Figure25A*), with cell ID 12209, 11572, 11585 and 12841 (*Figure25* from *B* to *E*).

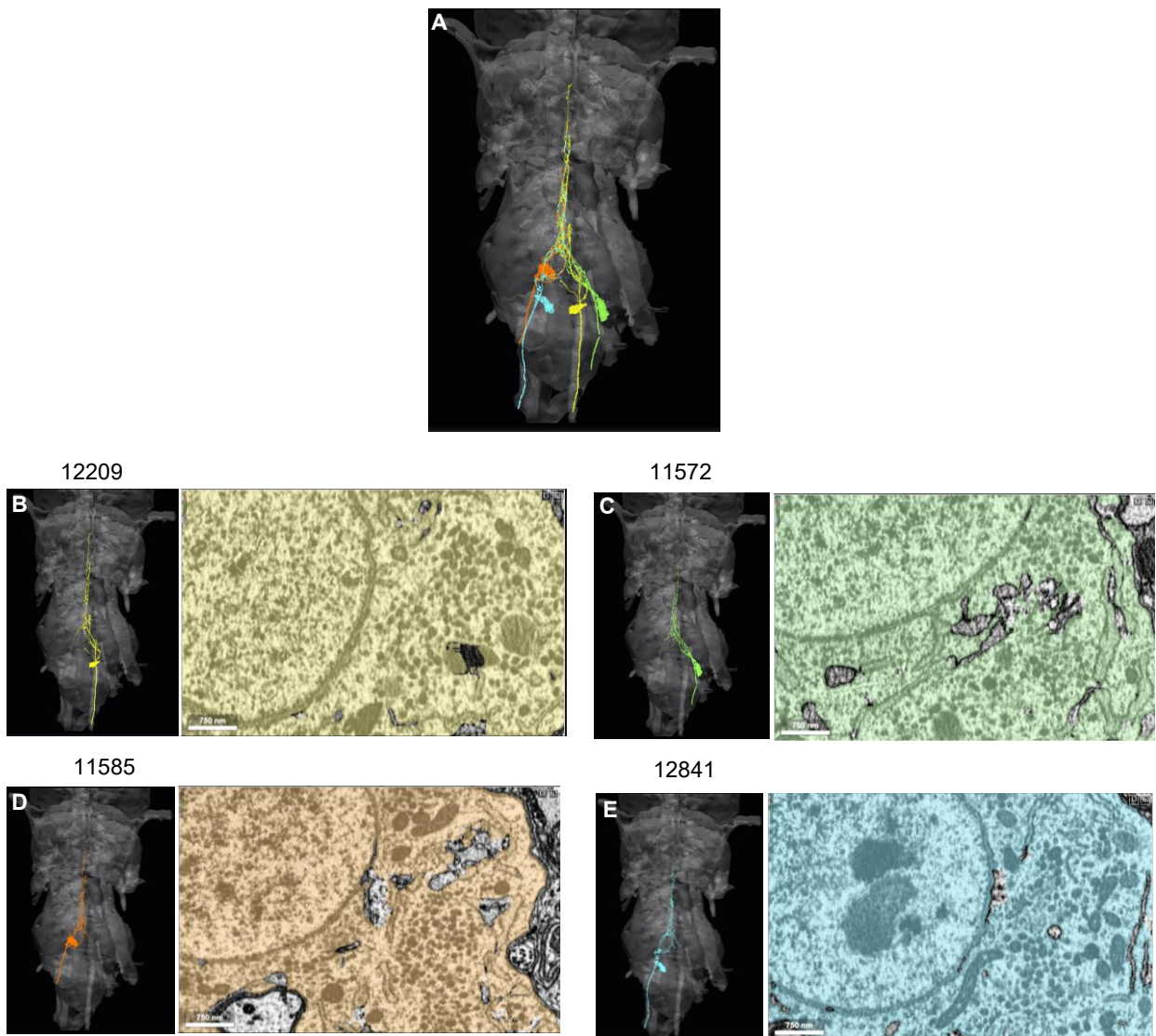


Figure 25: MNad18 type neurons. A. 3D view of neurons of MNad18. From B to E: 3D view of single neuron (on the left of each image), tangential section view at electron microscopy (on the right of each image) and along with corresponding cell ID. Scale bars: 750 nm.

MNad21 comprises of 4 neurons (*Figure26A*), with cell ID 10548, 10482, 10616 and 10473 (*Figure26* from *B* to *E*).

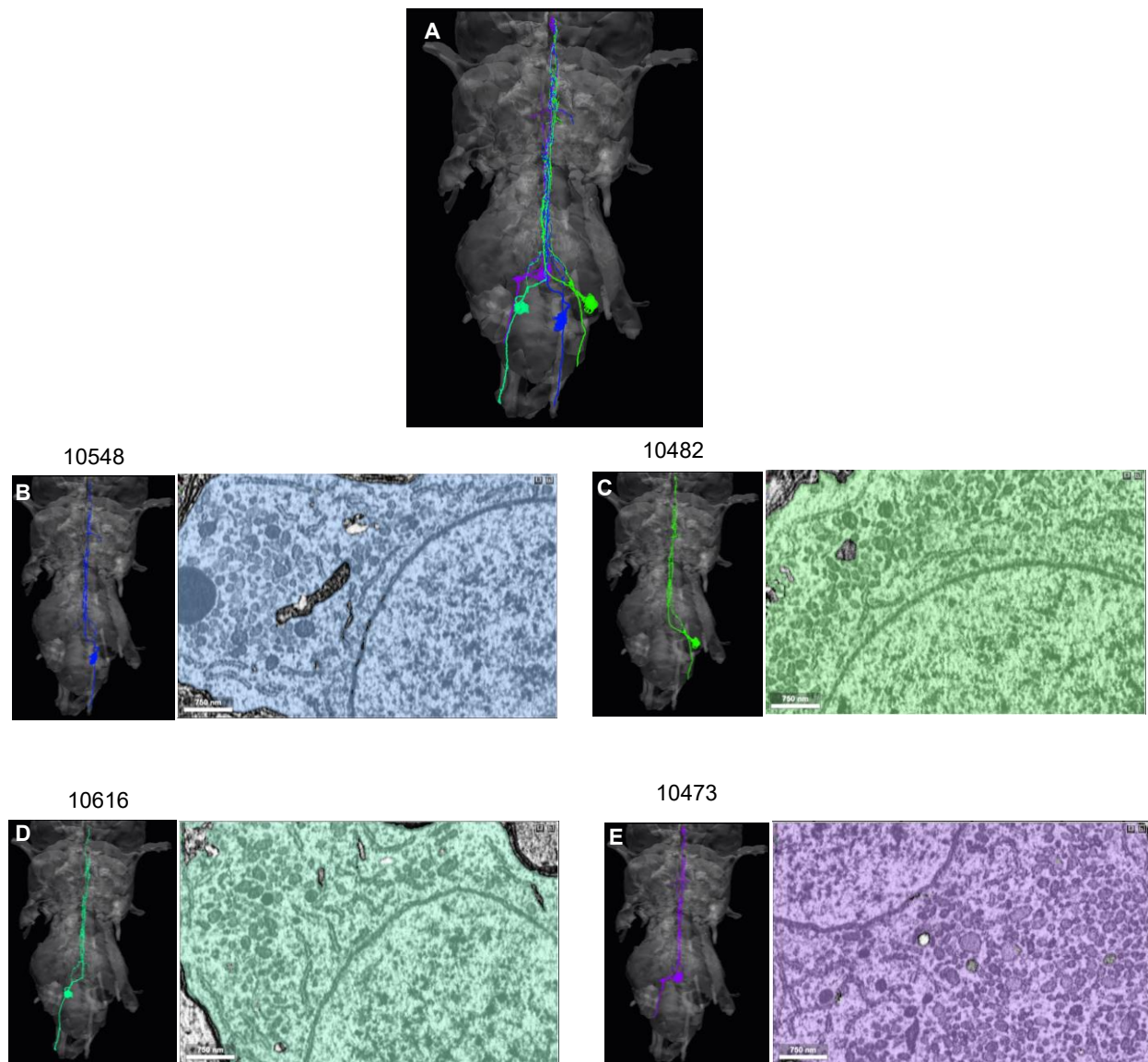


Figure 26: MNad21 type neurons. A. 3D view of neurons of MNad21. From B to E: 3D view of single neuron (on the left of each image), tangential section view at electron microscopy (on the right of each image) and along with corresponding cell ID. Scale bars: 750 nm.

MNad59 type consists of one neuron, whose cell ID is 13526 (*Figure27*).

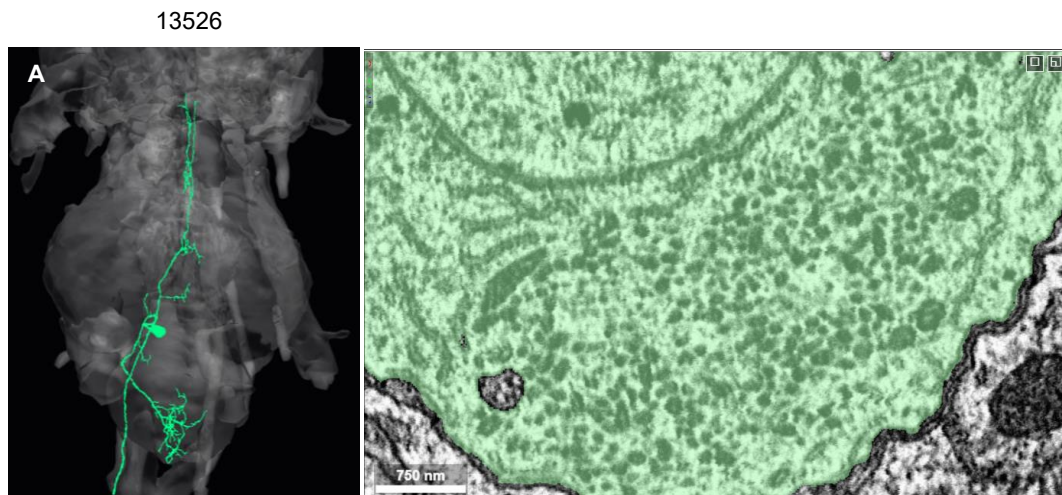


Figure 27: MNad 59 type neuron. A. 3D view of single neuron (on the left), tangential section view at electron microscopy (on the right) and along with corresponding cell ID. Scale bar: 750 nm.

MNad25 type consists of only two neurons (*Figure28A*), with cell ID 14370 and 13924 (*Figure28B* and C).

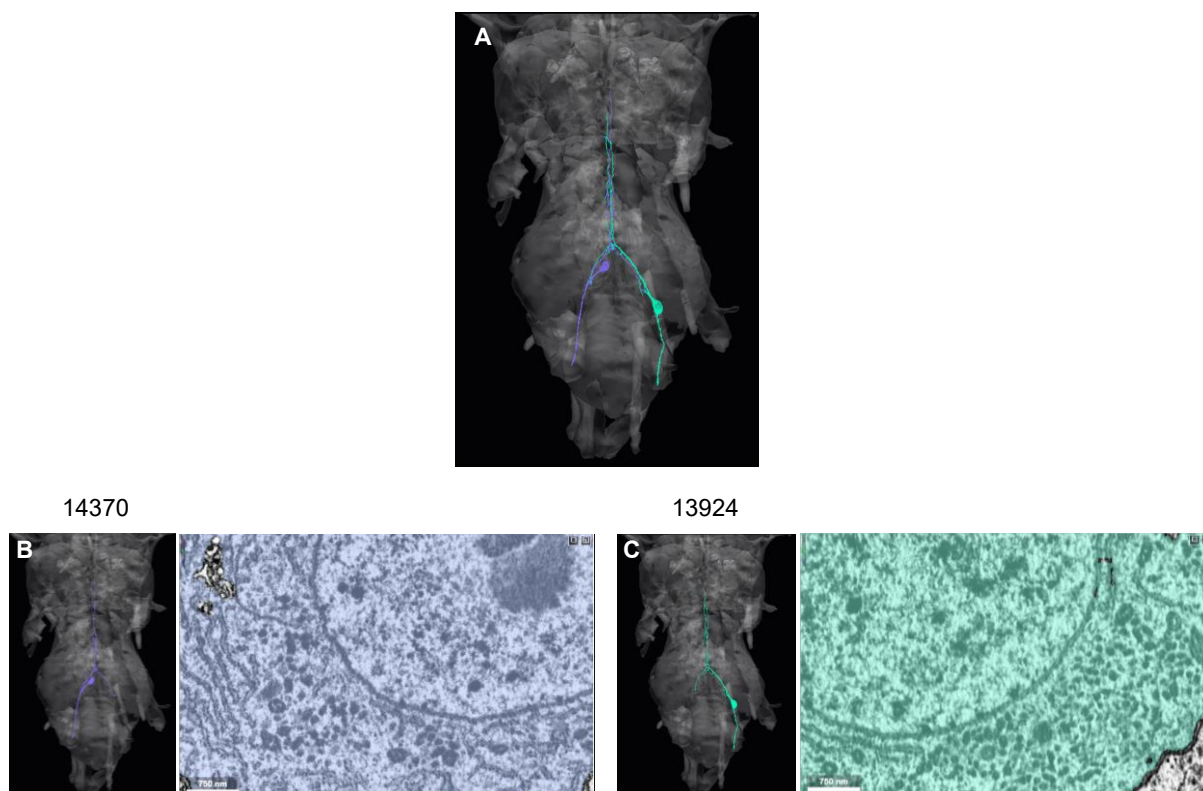


Figure 28: MNad25 type neurons. A. 3D view of neurons of MNad25. B and C: 3D view of single neuron (on the left of each image), tangential section view at electron microscopy (on the right of each image) and along with corresponding cell ID. Scale bars: 750 nm.

MNad52 type, also, consists of two neurons (*Figure29A*), their cell ID are: 14078 and 13929 (*Figure29B* and *C*).

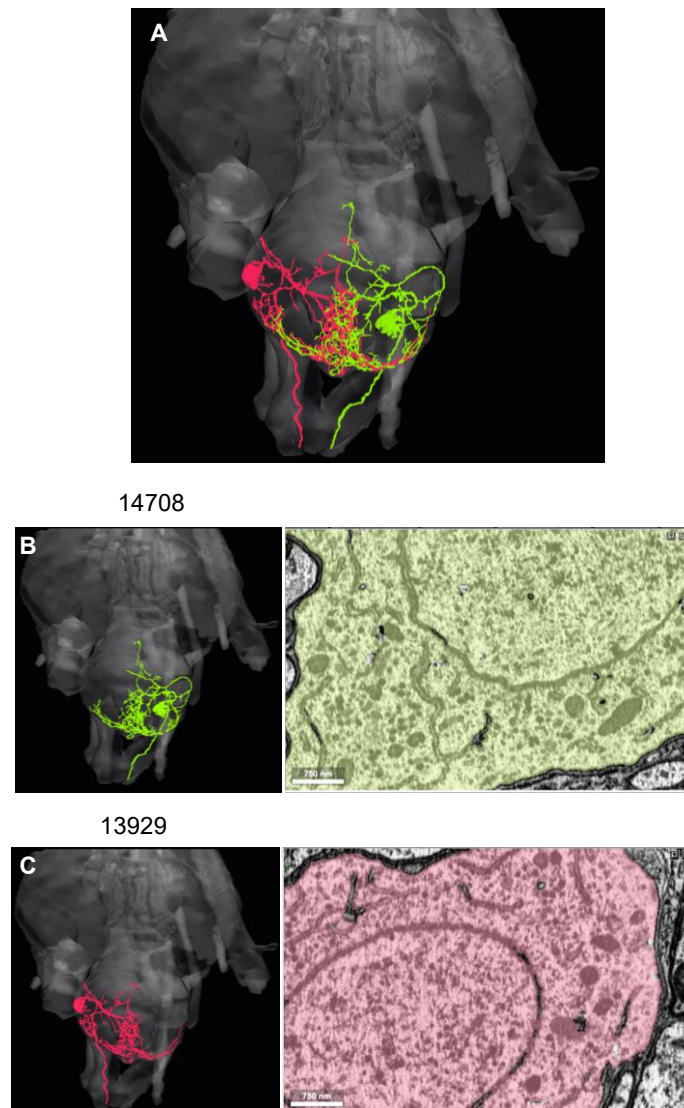
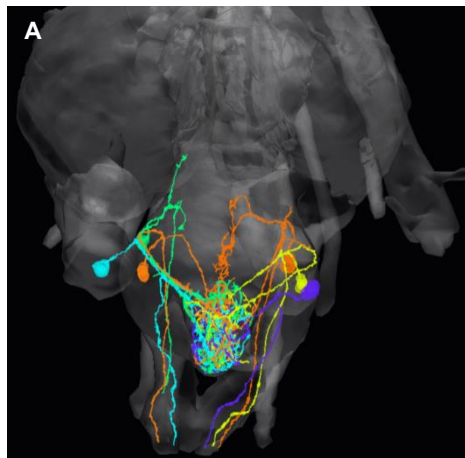
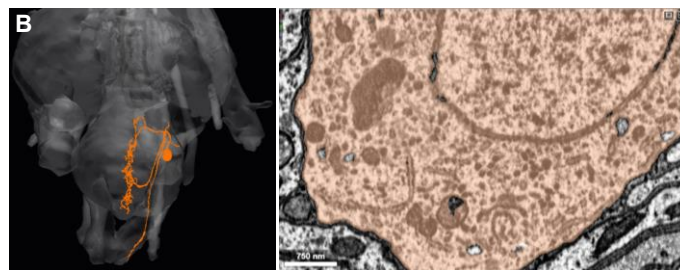


Figure 29: MNad52 type neurons. A. 3D view of neurons of MNad52. B and C: 3D view of single neuron (on the left), tangential section view at electron microscopy (on the right) and along with corresponding cell ID. Scale bars: 750 nm.

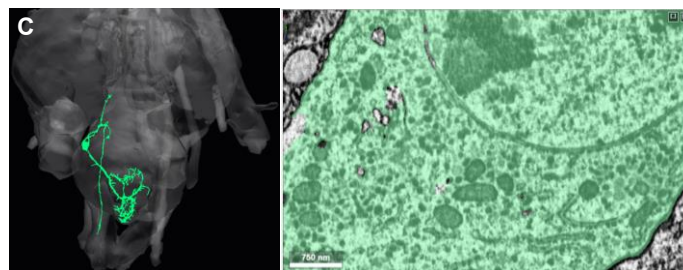
MNad03 type consists of six neurons (*Figure30A*), with cell ID 16305, 27988, 30702, 18060, 28921 and 16560 (*Figure30* from *B* to *G*).



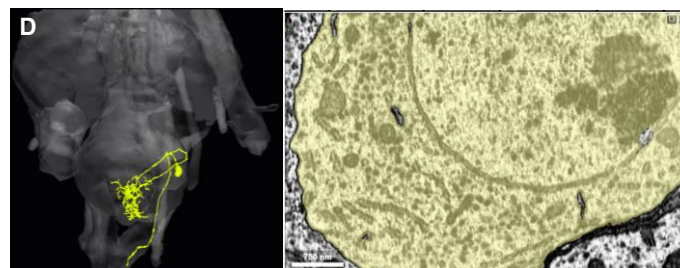
16305



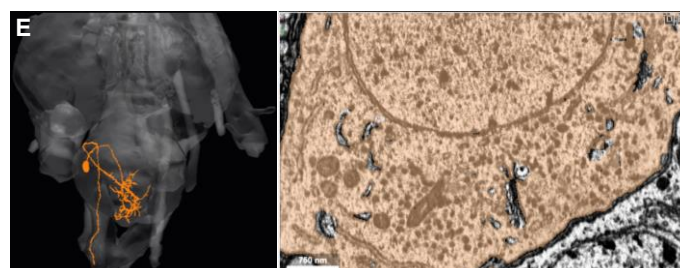
27988



30702



18060



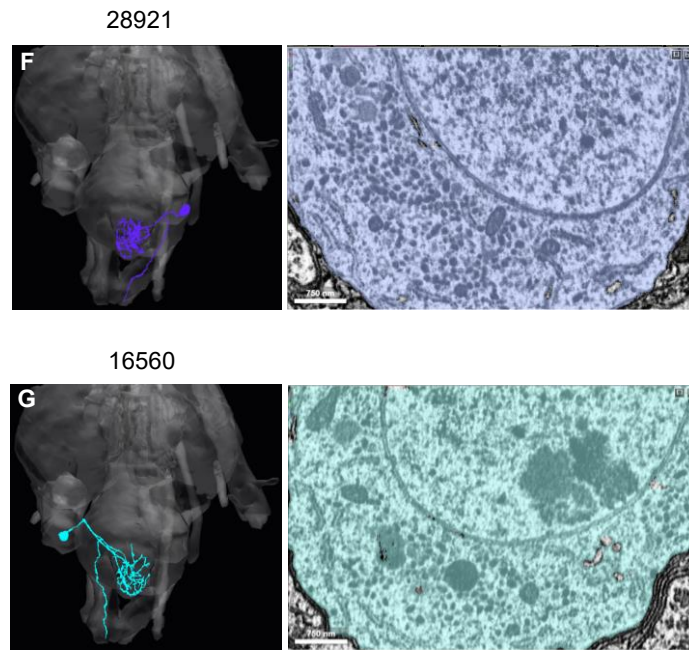
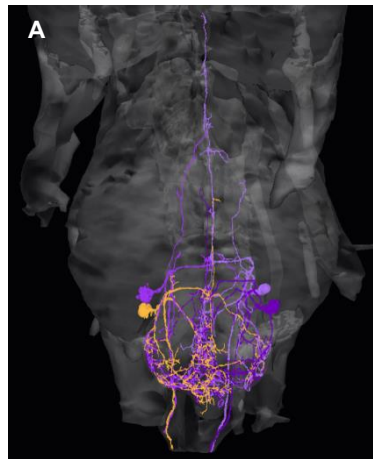
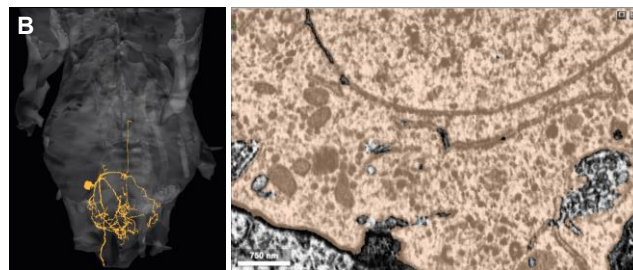


Figure 30: MNad03 type neurons. A. 3D view of neurons of MNad03. From B to G: 3D view of single neuron (on the left), tangential section view at electron microscopy (on the right) and along with corresponding cell ID. Scale bars: 750 nm.

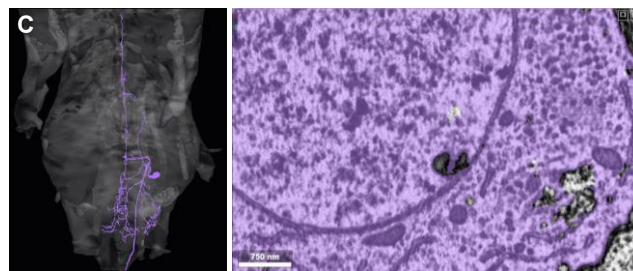
MNad07 type consists of 4 neurons (*Figure31A*), whose cell ID are: 22139, 13733, 14288 and 14768 (*Figure31* from *B* to *E*).



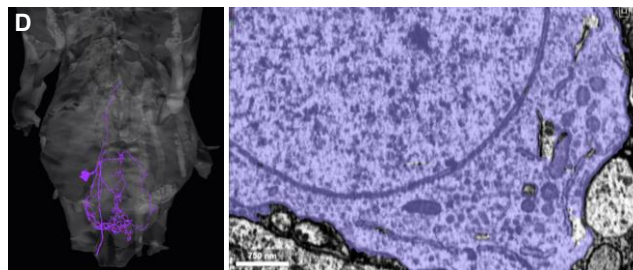
22139



13733



14288



14768

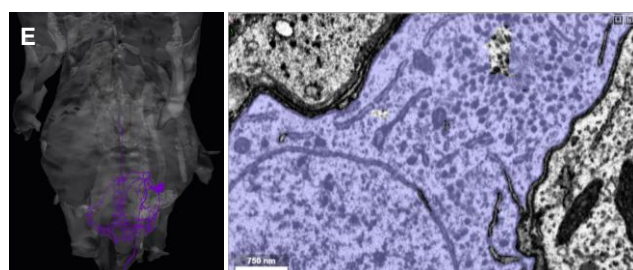


Figure 31: MNad07 type neurons. A. 3D view of neurons of MNad07. From B to E: 3D view of single neuron (on the left), tangential section view at electron microscopy (on the right) and along with corresponding cell ID. Scale bars: 750 nm.

Lastly, MNad37 type has only one neuron, whose cell ID is 12185 (*Figure32*).

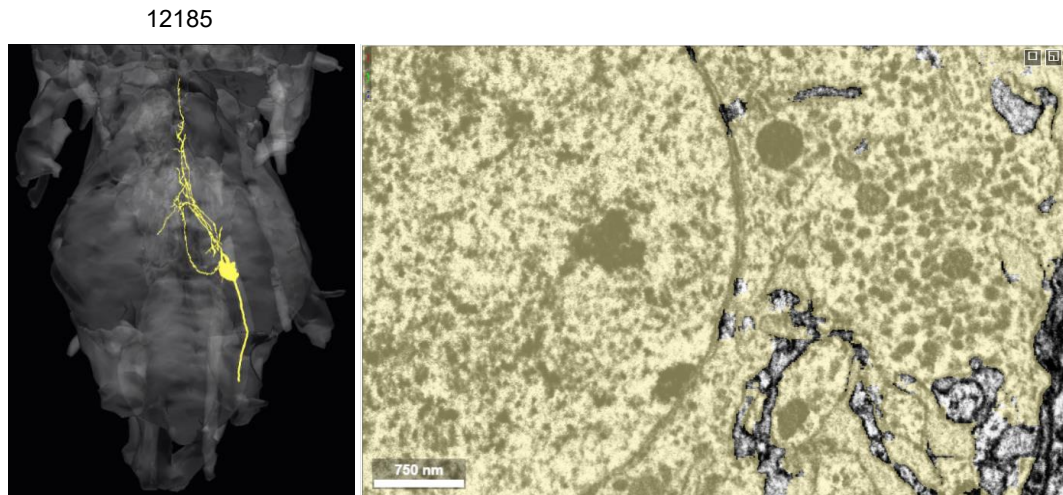


Figure 32: MNad37 type neuron. A. 3D view of single neuron (on the left), tangential section view at electron microscopy (on the right) and along with corresponding cell ID. Scale bar: 750 nm.

DISCUSSION

The aim of this thesis, my initial aim was to identify and functionally characterize neurohormonal receptors expressed in pericardial nephrocytes. Initial exploration of single-cell transcriptome analysis revealed the presence of several potential candidate receptors. Validation of the expression of these receptors was performed using the Gal4/UAS-system; employing the available Gal4 lines in stock, it was possible to establish the presence or absence of putative receptors.

Neurohormonal regulation of excretory system in both human and Drosophila

The excretory system is responsible for maintaining the appropriate osmotic balance and eliminating harmful substances, which is vital for the organism's survival. In humans, the renal system is subject to regulation by neurohormones. For example, the sympathetic nervous system can exert neural control over nephron function, affecting renal hemodynamics, excretory function, and secretion. Sympathetic nerve fibers release norepinephrine, which binds to receptors on renal blood vessels and nephrons. This can result in vasoconstriction of renal arterioles, reducing blood flow to the kidneys and decreasing urine production (Johns, 2014). Additionally, sympathetic stimulation can increase renin secretion from the juxtaglomerular cells of the kidney, initiating the renin-angiotensin-aldosterone system to raise blood pressure (Johns, 2014). Atrial natriuretic peptide is released by the heart in response to elevated blood volume and pressure. It acts on the renal tubules to promote sodium and water excretion while relaxing vascular smooth muscles, ultimately leading to a reduction in both blood volume and pressure (Song *et al.*, 2015).

Vasopressin, a peptide hormone with antidiuretic and antinatriuretic properties, can also influence renal function. It is produced by the hypothalamus and released by the posterior pituitary gland in response to increased blood osmolarity or decreased blood volume. Vasopressin acts on the collecting ducts of nephrons to enhance water reabsorption, thereby concentrating urine, conserving water, reducing urine output, and playing a pivotal role in maintaining body fluid balance and preventing dehydration (Kortenoeven *et al.*, 2015).

In *Drosophila*, the gene encoding a vasopressin-like peptide has been lost (Nässel and Zandawala, 2020); however, there exists a group of peptides with analogous roles: peptides DH31, DH44, and Capa promote fluid transport through their effects on principal cells ([Coast *et al.*, 2001](#); [Cabrero *et al.*, 2002](#); [Kean *et al.*, 2002](#)), whereas Lk peptide induces fluid secretion by targeting stellate cells ([O'Donnell *et al.*, 1998](#)).

The *Drosophila* excretory system consists of MTs and nephrocytes. MTs are analogous to the human kidney (Dow, 2009; Dow and Davies, 2006) and their function is hormonally regulated, involving by four major peptides (Capa-1, Lk, DH31, DH44) and one biogenic amine, tyramine

(Blumenthal, 2003). Receptors such as DH44R2 (Hector *et al.*, 2009), DH31R (Johnson *et al.*, 2005), and CAPAR (Terhzaz *et al.*, 2012) are expressed in principal cells, while stellate cells express the receptor for Lk (Radford *et al.*, 2002), demonstrating a role in regulating osmotic balance in *Drosophila*.

The other element within the excretory system is the nephrocytes, which are analogous to mammalian podocytes. These nephrocytes are capable of creating functional slit diaphragms through the utilization of orthologous proteins (Denholm and Skaer, 2009; Zhuang *et al.*, 2009, and Hermle *et al.*, 2017). Research focusing on the function of these cells, and even more crucially, their hormonal regulation, is currently lacking. Therefore, our study aimed address these knowledge gaps. Utilizing the Gal4/UAS system, the aim was to validate the existence of potential receptors for neurohormones, as suggested by the single-cell transcriptome analysis (Li *et al.*, 2022). This produced evidence supporting the presence of these receptors in nephrocytes, among these there are receptors that respond to biogenic amines, such as 5-HT7, which is activated by serotonin, others include AKHR and PDFR, which respond to neuropeptides. Additionally, DH31R and DH44R2 receptors were identified, and these receptors respond to neuropeptides with extensively documented effects on MTs. Furthermore, the presence of Gyc76C was observed, suggesting a potential role for this guanylyl cyclase in responding to neurohormonal regulation of these cells.

In the future, it would be interesting to explore whether similar neurohormonal regulatory mechanisms exist in higher vertebrates, at various levels of function.

Serotonin but not octopamine influences intracellular Ca^{2+} concentration in nephrocytes

To explore the feasibility of tracking nephrocyte activity when exposed to various neurohormones, this thesis aimed to investigate calcium signaling through the utilization of calcium imaging techniques. These measurements were conducted on live samples incubated with ligands, which were presumed to either trigger or inhibit calcium-mediated signal transduction within the cells. It is worth noting that were encountered challenges when attempting to use the GCaMP sensor for measurements. This was primarily due to the difficulty in maintaining the stability of pericardial nephrocytes, as they are attached to the *Drosophila* cardiac tube, which continued to beat even after the sample was dissected in HLS, making measurements impractical. Therefore, it was employed the CaLexA system, which induces GFP expression in response to a rise in calcium levels (Masuyama *et al.*, 2012). However, the adoption of this system has restricted the options when it comes to selecting compounds for testing the cellular response. The primary constraint lies in the system's dependence on intracellular Ca^{2+} concentration, while many

neuropeptides exert their effects by modulating cAMP levels (Harmar, 2001; Hector *et al.*, 2009; Mertens *et al.*, 2005).

The decision to utilize biogenic amines, as serotonin and octopamine, into the experiment was based on their established effects, because both have the capability to induce an increase in intracellular Ca^{2+} concentration (Sampson *et al.*, 2020; Balfanz *et al.*, 2005; Roeder, 1999). The observed reduction in fluorescence intensity in serotonin-treated pericardial nephrocytes can be attributed to a decrease in intracellular Ca^{2+} concentration within these cells. This experiment allows us to explore the collective impact of various 5-HT receptors. Through single-cell transcriptome analysis revealed the presence of three distinct serotonin receptor transcripts: 5-HT7, 5-HT2A, and 5-HT2B. While 5-HT7 has been associated with increased cAMP levels (Rosikon *et al.*, 2023) and demonstrated in various studies, including cultured cells (Witz *et al.*, 1990; Saudou *et al.*, 1992) and in ellipsoid body cells of *Drosophila* (Liu *et al.*, 2019), it seems unlikely to be the primary mediator in this context. Regarding the other two receptors, 5-HT2A and 5-HT2B, the intracellular signaling pathway for these 5-HT2 receptors in *Drosophila* remains unexplored. Recent findings, however, indicate that 5-HT2B can lead to significant elevations in baseline intracellular calcium levels in lamina monopolar L2 neurons of *Drosophila* (Sampson *et al.*, 2020).

In mammals, the 5-HT2 ortholog receptors constitute a receptor family that is linked to the $G_{q/11}$ pathway (Hannon and Hoyer, 2008). The $G_{q/11}$ family of heterotrimeric G proteins triggers the activation of β -isoforms of phospholipase C, leading to the release of calcium ions from intracellular stores and the activation of protein kinase C (Exton, 1996).

Further studies should be addressed to evaluate contribution of each individual receptor appropriately in response to serotonin to better clarify this effect. It should be interesting also to investigate their intracellular signaling pathways, conducting receptor knockdown experiments, mutagenesis studies on the receptors or their G-protein coupled, and potentially employing antagonists. In particular a dose-dependent study could help to identify receptor saturation and to determine the minimum concentration of serotonin able to activate nephrocytes.

Thus, the observed data appears to suggest a baseline calcium activity within these cells, which is seemingly altered when serotonin is introduced, resulting in a decrease in transcription levels. To validate these findings, it is necessary to replicate this experiment using alternative methods, since visualize inhibition with CaLexA can be challenging due the fact that there is not removal of generated GFP.

From analysis of single-cell transcriptomes, it revealed the presence of 2 distinct octopamine receptor transcripts, specifically Oct α 2R and Oct β 2R. Previously, through Gal4/UAS screening validation, it was already established that the Oct α 2R receptor was not expressed in nephrocytes. Since Oct α 2R expression is based on single-cell transcriptome data, there may be alternate isoforms of this receptor that were not expressed by the Gal4 line used in the experiments. Alternatively, there could be the presence of Oct β 2R expression in nephrocytes. On this idea, these cells underwent octopamine exposure. However, nephrocytes treated with octopamine shown no discernible difference in fluorescence intensity compared to control, indicating that it had no effect on the intracellular calcium concentration in these cells. The activation of these receptors mediated by octopamine initiates signal transduction pathways, which could either regulate intracellular cAMP concentrations or result in longer-term effects, such as those induced via the activation of transcription.

Nonetheless, the reliability of single-cell transcriptome data was demonstrated since it was independently verified the presence of some of the expected receptors using the Gal4/UAS system.

Excretion process unaffected by RNAi of receptors

The formation of primary urine is a process that involves secretion by MTs and is also regulated by neurohormones (Davies *et al.*, 1997; O'Donnell *et al.*, 1998; Coast *et al.*, 2001; Cabrero *et al.*, 2002; Blumenthal, 2003). This primary urine can mix with digestive tract contents and move toward the hindgut. Essential metabolites, like ions and water, are selectively reabsorbed mainly in the ileum and rectum, influencing the final composition of excreted waste products (Gäde, 2004). Osmotic hemolymph homeostasis in insects depends on their ability to control water and ion levels. As nephrocytes play a significant role in regulating hemolymph composition, the aim was to investigate whether knocking down the receptors for these neurohormones would result in any alterations in excretion. To assess this, it was monitored defecation rates over a time window of 2, 4 and 6 hours. The knockdown of receptors did not reveal any significant alterations in the frequency of defecation events when compared to the two control groups (*Figure 18* from *A* to *E*). This phenotype was consistent even in positive control *hid,rpr*, where nephrocytes were expected to be dead (*Figure 18F*). These findings suggest that these receptors may not play a significant role in the regulation of the nephrocyte-mediated excretion process. It is important to acknowledge that the quantification of defecation events relied on manual counting of blue dots, which introduces the possibility of both underestimation and overestimation in our estimates. In addition, given the circular shape of the tubes, it may become challenging to avoid potential errors during the count, such as the inadvertent double-counting or slip-up of points. Furthermore, since the experimental crosses were maintained in an incubator at 29°C to relieve Gal80 repression, it is plausible that

some of the blue dots may have evaporated or shown varying color textures before being counted, making them challenging to distinguish; particularly given the inability to precisely control humidity conditions.

In the future, the significance of these receptors could be evaluated through alternative assays, such as assessing starvation resistance, feeding or thirst behaviors, and investigating desiccation tolerance as well as metabolism since there is a crosstalk between nephrocytes and fat body, the primary tissue for storing energy and involved in metabolism (Canavoso *et al.*, 2001).

The complex network of interactions regulating metabolism in Drosophila

In *Drosophila*, the interaction between the fat body and nephrocytes assumes a pivotal role in the regulation of metabolism and overall homeostasis. This interplay encompasses nutrient exchange, where the fat body serves as a store for nutrients, predominantly lipids and glycogen, and can release them into the hemolymph when needed (Arrese and Soulages, 2010). Furthermore, this intricate interconnection extends to the gut, in which enteroendocrine cells capable of sensing the internal intestinal environment and secreting neurohormones that modulate various physiological processes, including gut motility, appetite, and nutrient homeostasis (Miguel-Aliaga *et al.*, 2018). Notably, tachykinins derived from the gut are instrumental in regulating intestinal lipid metabolism (Song *et al.*, 2014). Single-cell transcriptome analysis revealed the presence of transcript for the tachykinin receptors in nephrocytes, specifically TkR99D (Tachykinin-like receptor at 99D) (*Figure 8*). This suggests the possibility that peptides also originating from the gut, may have a potential influence on nephrocyte functions.

Another peptide involved in the regulation of fat metabolism is the Adipokinetic hormone (AKH), instead, synthesized within the neuroendocrine cells of the corpora cardiaca (Nässel and Zandawala, 2019). The AKH receptor was extensively studied, and it is well-documented its role in both fat storage and mobilization (Bharucha *et al.*, 2008). Consequently, the knockdown of this receptor in nephrocytes may disrupt the delicate balance of communication between the fat body, nephrocytes, and the gut. This disruption may shift to fat accumulation, resulting in an increase in fresh weight (*Figure 19A*). Similarly, the knockdown of the Gyc76C receptor (*Figure 19E*) also suggests its potential involvement in the regulation of metabolism, as indicated by the observed increase in fresh weight.

Apart from their primary role as a barrier, pericardial nephrocytes are responsible for producing Upd3, a cytokine that signals to the fat body. This interaction plays a crucial role in regulating the specific expression of Pericardin, a protein of extracellular ECM that is essential for maintaining proper cardiac function (Gera *et al.*, 2022). When it was interfered with the receptors

by employing dotGal4, it could potentially disrupt the release of Upd3, subsequently influencing the storage of nutrients. Consequently, alterations in nutrient stores may lead to variations in dry weight, as experiments of AKH- and DH44 receptor knock down have shown (*Figure19 A and C*).

Knocking down the AKH and Gyc76c receptors in nephrocytes resulted in an increase in water content (*Figure19 A and C*). This shift in water balance can be attributed to several factors, including reduced water loss and potentially increased water intake by the flies. Furthermore, it may stem from alterations in the regulation of homeostasis, as the gut also plays a pivotal role in maintaining ion balance and water absorption in *Drosophila* (Miguel-Aliaga *et al.*, 2018), working in conjunction with the MTs (Jung *et al.*, 2005; Dow *et al.*, 2010; Miller *et al.*, 2013).

Consistent results were observed in hid,rpr genotype (*Figure19F*), in which was induced nephrocyte cell death. It is important to highlight that, due to the severity of the resulting phenotype, only two replicates of this experiment could be conducted, as the survival of the flies was compromised.

These findings suggest that perturbing the delicate balance involved in sustaining homeostasis in *Drosophila*, due to shifts in the interactions between various tissues and cell types, ultimately resulting in changes in both weight and water content.

Identification of putative neurons producing Lk and DH44 in Drosophila VNC

The cell bodies of neurosecretory cells within the adult VNC are primarily located in the abdominal neuromeres, with only a set of thoracic neuromeres containing them. In the abdominal neuromeres, these cells produce peptide hormones that play a role in regulating water and ion balance. Among these hormones are LK, DH44, CAPA1, CAPA2, and DH31 peptides (Nässel and Zandawala, 2020). The receptors for DH44, CAPA1 and DH31 are expressed in principal cells, while the LK receptor is found in stellate cells within the MTs. This distribution suggests that these peptides are involved in the regulation of ion transport (Nässel and Zandawala, 2019). The identification of neurohormone-producing neurons influencing the MTs in the VNC of an adult female was performed using *neuPrint+*. Recent progress in electron microscopy and deep learning has made it feasible to reconstruct a connectome, which provides a comprehensive description of neurons and their interconnections, even for extensive neural tissue volumes such as VNC.

This enabled us to potentially pinpoint neurons capable of producing Lk and DH44 neurohormones, as some Lk neurons were found to co-express DH44 (Zandawala *et al.*, 2018). The identification process involved searching for characteristic features of neurohormone-producing neurons, such as large cell bodies containing dense core vesicles. During this analysis,

neurons exhibiting the morphology similar to that observed through confocal microscopy were categorized as putative neurons (neurons of MNad13, MNad38, MNad27, MNad18, MNad21, MNad59, MNad25, MNad52, MNad03, MNad07 types), based on their overall shape that resembles a violin box (Zandawala *et al.*, 2018). It is important to acknowledge the limitations of this identification method, as it assumes that the identified neurons are indeed responsible for producing neuropeptides Lk and DH44. Furthermore, the search was conducted manually within the abdominal section of the VNC, potentially leading to an underestimation of the total number of identified neurons. Additionally, this identification was carried out in adult female *Drosophila*, prompting the need for future evaluations in adult male to assess potential sex-based differences. Moreover, it's crucial to recognize that the connectome data provides a static snapshot, representing the conditions at the time of analysis of the samples, which may not accurately reflect normal physiological variations that can occur between individuals, throughout the day, or in response to different environmental conditions.

However, despite these limitations, this identification process lays the foundation for further research and insights into the regulation of these neurons through synaptic interactions and their interplay with one another. This will contribute significantly to a deeper understanding of how the excretory system in *Drosophila* is modulated by these neurohormones.

CONCLUSION

In this thesis, attempt was made to improve the existing gaps in our understanding of the regulation of pericardial nephrocytes. These cells, owing to their similarities with human podocytes, are gaining prominence as a valuable model for investigating renal diseases, notably glomerulopathies.

A better understand of neurohormonal regulation of pericardiac nephrocytes could help to identify new therapeutic avenues. The relative simplicity of the *Drosophila* excretory system and the presence of neurohormones homologous to those in humans make this an intriguing avenue for exploration, with the promise of innovative therapies.

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